



# JoinUp

## Group members:

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## **Acknowledgement:**

We the team behind this project extend our heartfelt gratitude to Allah followed by our parents whose unwavering support has been our foundation We also express our sincere appreciation to our esteemed professors Dr. Alyaa and Dr. Ashwaq for their invaluable guidance and dedication in educating and mentoring us Lastly we take a moment to acknowledge and appreciate our own efforts commitment and perseverance throughout this journey

## **Chapter 1 (Business requirements):**

### **Application type:**

Interactive transaction-based

### **Brief Description:**

The application serves as a unified platform that enables students to easily register for and participate in university activities and events. It streamlines the organizational process, enhances communication within the university community, and provides real-time updates and reminders for important events.

### **Problem it Solves:**

activity dates and This application tackles a prevalent issue among students: difficulty in keeping track of various deadlines related to university events.

### **Target Users:**

University students across different majors.

Administrators and event organizers responsible for managing and coordinating university events

## **Chapter 2 ( Stakeholders):**

### **Stakeholders:**

- 1- University Students: These are the primary users of the application. They will use the platform to register for events, track their activities, and interact with the community.
- 2- Event Manager : manage the events, send out invitations, collect feedback, and maintain communication with the participants.
- 3- University Administrators: These stakeholders oversee the platform, ensuring its smooth operation, implementing any required updates, and addressing issues related to the app's functionality and user experience.

### **Requirements Gathering Methods:**

Online surveys

## **User Interviews:**

### **Student 1:**

#### **Information:**

- **Name:** Sarah Al-Rifai
- **Email:** [s443004961@uqu.edu.sa](mailto:s443004961@uqu.edu.sa)
- **Mobile:** +966 57 195 7836

#### **Questions and Answers:**

- **Do you think having an app that provides all event information in one place would save you time and effort?**  
"Yes, it would help a lot."  
"I think so. I won't have to spend much time checking for university updates and events."
- **What features would you like to see in the app?**  
"Tracking members' progress in activities, a section for suggesting new event ideas, sending notifications for important updates, supporting event feedback after it ends, and offering direct contact with organizers."

### **Student 2:**

#### **Information:**

- **Name:** Shatha Al-Omari
- **Email:** [s445005964@uqu.edu.sa](mailto:s445005964@uqu.edu.sa)
- **Mobile:** +966 55 159 5737

#### **Questions and Answers:**

- **Do you think easy access to event information and club registration dates would improve your university experience?**  
"Yes."
- **How do you expect the app to impact your interaction with activities and ease of access to them?**  
"It would definitely make it easier to stay informed about all events."  
"To some extent, it will help me keep up with all the news."

**Student 3:**

#### **Information:**

- **Name:** Dana Al-Sarihi
- **Email:** [s444000911@uqu.edu.sa](mailto:s444000911@uqu.edu.sa)
- **Mobile:** +966 56 001 6230

#### **Questions and Answers:**

- **Do you think having an app that provides all event information in one place would save you time and effort?**

“Yes, it would be very helpful. I’ll stay connected with all the college interests and activities, so having an app gather all this info would be an excellent option.”

- **What features would you like to see in the app?**

“Yes, it would be very helpful. I’ll stay connected with all the college interests and activities, so having an app gather all this info would be an excellent option.”

#### **Student 4:**

##### **Information:**

- **Name:** Layan Al-Rahili
- **Email:** [s444004332@uqu.edu.sa](mailto:s444004332@uqu.edu.sa)
- **Mobile:** +966 54 264 9090

##### **Questions and Answers:**

- **Do you think easy access to event information and club registration dates would improve your university experience?**

“Yes.”

- **How do you expect the app to impact your interaction with activities and ease of access to them?**

“I think it will help a lot, especially for students who don’t use social media much. Having an app will be useful.”

#### **Manager:**

##### **Information:**

- **Name:** Layan Al-Rahili
- **Email:** [s445007749@uqu.edu.sa](mailto:s445007749@uqu.edu.sa)
- **Mobile:** +966 50 245 3253

##### **Questions and Answers:**

- **What are the biggest challenges you currently face in managing the club's activities?**

“Time management + task delegation + forgetting events.”

- **Do you want a unified system to organize all events and activities easily?** “Yes.”

- **How do you currently schedule events and communicate with club members?** “Via WhatsApp and the Planner app.”

- **Do you think providing a platform that facilitates student registration for events will increase participation?**

“Yes, and it will allow for expansion, better awareness of important activities and events, as well as membership registration and renewal.”

### Chapter 3: Glossary:

| Glossary                 |                                                                                                                                          |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Word                     | Meaning                                                                                                                                  |
| Event Registration       | The process by which students sign up to participate in an event or activity via the application.                                        |
| User Profile             | The personal information of a user, such as name, contact details, and preferences stored in the system.                                 |
| Activity Calendar        | A calendar displaying all upcoming events and activities that the user is registered for, with details such as date, time, and location. |
| Notification Preferences | The settings that allow users to choose how and when they are notified about event updates or reminders.                                 |
| Feedback                 | The responses or reviews provided by students about an event after it has taken place.                                                   |
| Admin Dashboard          | The interface which used by event organizers or administrators to manage events, announcements, and user data.                           |
| Session Timeout          | The period after which a user session will expire if inactive for a defined time.                                                        |

## Chapter 4 ( Software Requirements):

### Functional requirements:

#### 1- The user should be able to log in.

- 1.1 The system should check the format of the username and password.
- 1.2 If the format is incorrect the system should display an error message specifying the issue.
- 1.3 The system should check the database if the entered data is existed.
- 1.4 If the username data does not exist it should display an error message.
- 1.5 If the username does exist the system should compare the stored password to the entered password.
- 1.6 If the password does not match, the system should display an error message.
- 1.7 The system should check the authorization by email domain

#### 2- The user should be able to register for activities.

- 2.1 The system should check if the username format is correct and does not contain invalid characters.
- 2.2 If the format is incorrect, the system should display an error message .
- 2.3 The system should search the event registration database to see if the username is already registered.
- 2.4 If the username exists, the system should display a message saying the user is already registered and prevent duplicate registration.
- 2.5 If the username does not exist, the system should continue with the registration process.
- 2.6 The system should check if the event is still open for registration.
- 2.7 If registration is closed, the system should inform the user that they cannot register.
- 2.8 If registration is open, the system should add the user's information to the event database.
- 2.9 The system should show a confirmation message once the registration is successful.

#### 3- The user should have a calendar for registered activities.

- 3.1 The system should display events that match the calendar time.
- 3.2 The system should show event details based on the time selected by the user.
- 3.3 The system should support multiple calendar views (daily, weekly, monthly).

#### 4- The user should receive optional notifications about activities.

- 4.1 The system should send optional notifications via email or in-app alerts.
- 4.2 The system should allow users to customize notification preferences or opt out.
- 4.3 The system should ensure reliable notification delivery with retries.
- 4.4 The system should maintain a history of received notifications.

#### 5- The user should be able to handle announcements.

- 5.1 The system should allow managers to compose announcements with formatting and attachments.



- 5.2 The system should display announcements on the dashboard or send them via email.
- 5.3 The system should enable scheduling of announcements for specific times.
- 5.4 The system should maintain a log of announcements for tracking.

6- The user should provide feedback on events.

- 6.1 The system should verify if the user is registered for the event.
- 6.2 If registered, the system should display a feedback form.
- 6.3 The system should allow users to submit feedback.
- 6.4 The system should store the submitted feedback in the database.
- 6.5 The system should display a confirmation message after successful feedback submission

## 7- The user should view their event history.

- 7.1 The system should retrieve all registered events from the database.
- 7.2 The system should calculate and display the total number of attended events.
- 7.3 The system should show the user's participation frequency and feedback provided.
- 7.4 If no events are attended, the system should display a message indicating no data is available.
- 7.5 The system should provide a graphical representation of event participation.

## 8- The user should see event locations on Google Maps.

- 8.1 The system should integrate with Google Maps to display event locations.
- 8.2 The system should show directions to the event venue.
- 8.3 The system should provide navigation options, including routes and transportation.
- 8.4 The system should display additional location details, such as parking or transport routes.

## 9- The user should cancel activity registrations.

- 9.1 The system should provide a "Cancel Registration" option.
- 9.2 The system should check if the event allows cancellations.
- 9.3 If cancellation is not allowed, the system should display a message.
- 9.4 The system should remove the user from the event's participant list upon successful cancellation.
- 9.5 The system should notify the user of successful cancellation.
- 9.6 The system should update the user's calendar to remove the canceled event.
- 9.7 If the event has limited capacity, the system should make the spot available for others.

## 10- The user should register as a club member.

- 10.1 The system should validate input data (valid email and secure password).
- 10.2 The system should check for duplicate accounts.
- 10.3 The system should encrypt passwords before storing them.
- 10.4 The system should securely save user data in the database.
- 10.5 The system should send a confirmation email upon registration.
- 10.6 The system should allow users to modify profile details post-registration.
- 10.7 The system should enable password recovery via email.
- 10.8 The system should log all registration activities for auditing.

## 11-The user should complete email verification after logging in.

- 11.1 The system should send a verification code to the user's registered email.
- 11.2 The system should request the user to enter the verification code.
- 11.3 The system should validate the entered code against the one sent to the email.
- 11.4 If the code is incorrect, the system should display an error message and allow the user to retry.
- 11.5 If the code is correct, the system should grant the user full access to their account.

## **Non-Functional:**

### **1- Performance:**

- 1.1 - The account creation process handles user data within 3-5 seconds using scalable algorithms with secure data verification.
- 1.2 - The login process completes authentication within 2-3 seconds, leveraging caching techniques without compromising security.
- 1.3 Search results for university activities are displayed within 3 seconds, ensuring data is protected during processing and transmission.

### **2- Security:**

- 2.1 - Passwords are encrypted using strong hashing techniques (e.g., SHA-256 with salting), and all sensitive data is stored in an encrypted format.
- 2.2 - All user inputs are validated and sanitized to prevent SQL Injection and XSS attacks, with continuous monitoring to detect abnormal user behavior.
- 2.3 User sessions are managed with unique, encrypted session identifiers, and inactive sessions expire automatically after 15 minutes.
- 2.4 - All data transmitted between users and the system is encrypted using HTTPS with up-to-date SSL certificates.

### **3- Reliability:**

- 3.1 The system should maintain a failure rate below 1%.
- 3.2 The system should handle 100 concurrent requests efficiently.
- 3.3 Calendars should update within 5 seconds after registration.
- 3.4 Error messages should provide general feedback, with detailed logs for admins.
- 3.5 The system should prevent duplicate or unauthorized submissions.
- 3.6 Notifications should process at 5,000 per second under load.
- 3.7 Feedback operations should complete within 3 seconds.

### **4- Maintainability:**

- 4.1 - The system allows administrators to deploy updates or security patches without any downtime (Zero Downtime Deployment).
- 4.2 - Detailed system logs are securely stored for 1 year and can be easily accessed for troubleshooting or audits.
- 4.3 - All new features are designed to integrate seamlessly with the existing system within 2 weeks, without impacting performance or security.

### **5- Availability:**

- 5.1 - Critical updates (e.g., new events or schedule changes) are propagated across all systems within 5 seconds using reliable distribution mechanisms.
- 5.2 - An automatic failover mechanism ensures service continuity during hardware or software failures, with recovery time not exceeding 30 seconds.

5.3 - The system is designed to maintain 99.99% uptime annually, with contingency plans for any unexpected downtime.

#### 6- Usability:

6.1 - The system ensures full compatibility with mobile and desktop devices with an intuitive user interface.

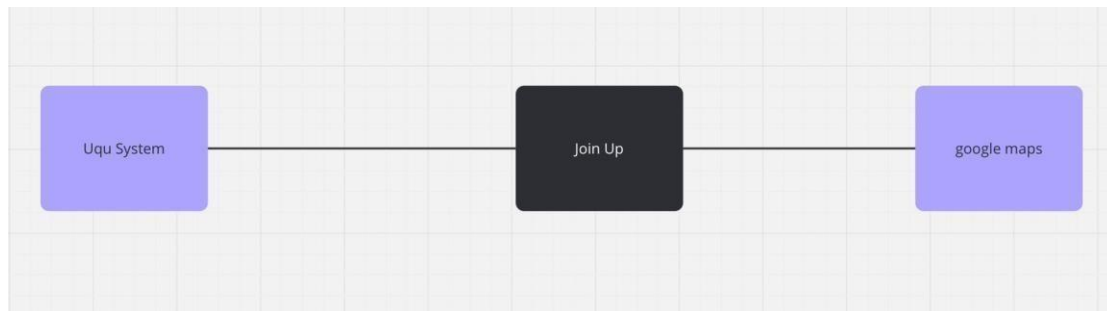
6.2 - Core tasks like registration and feedback submission are completed within 3 clicks.

6.3 - A dark mode option is available across all pages to enhance user experience.

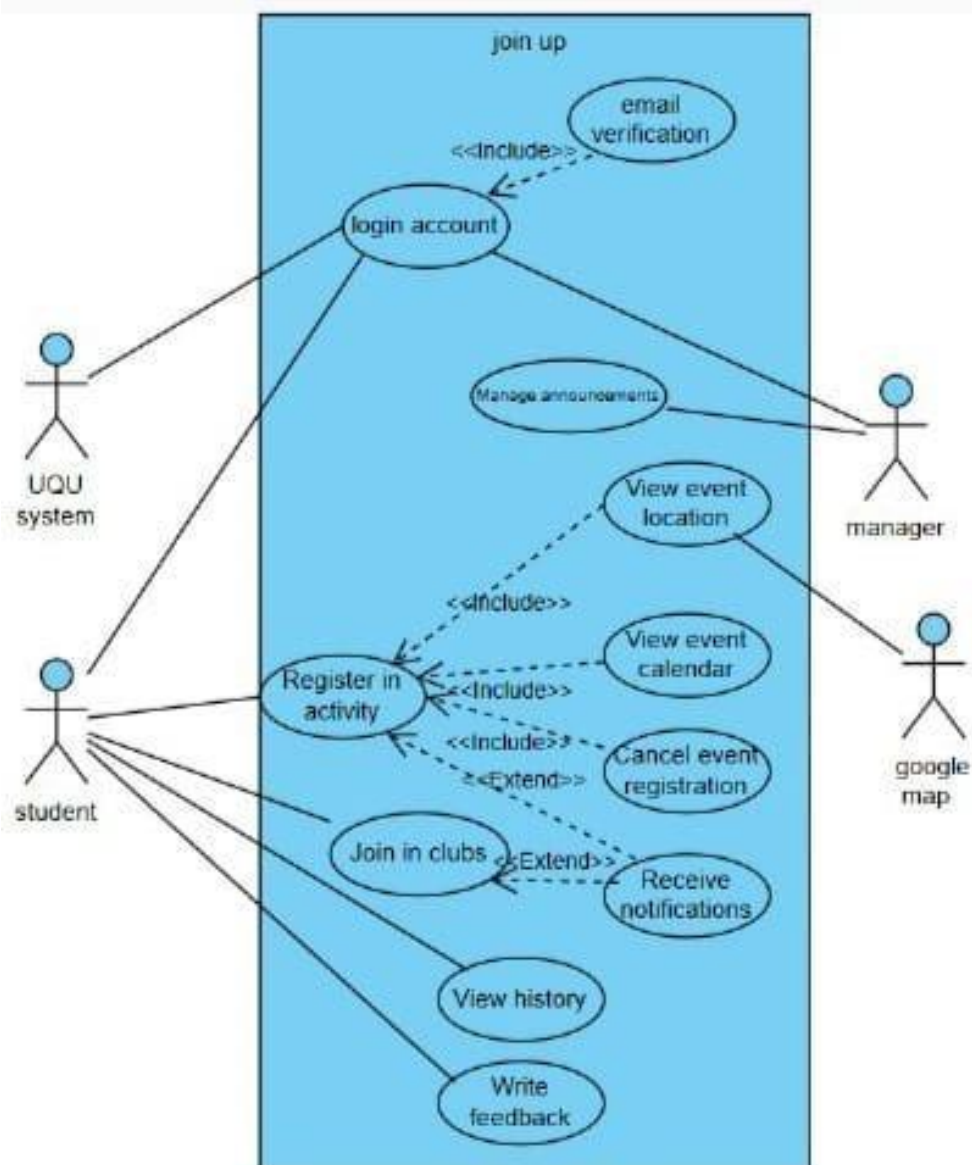
6.4 - Error messages clearly explain the problem and provide actionable suggestions for resolution.

## Chapter 5 (Software Modeling):

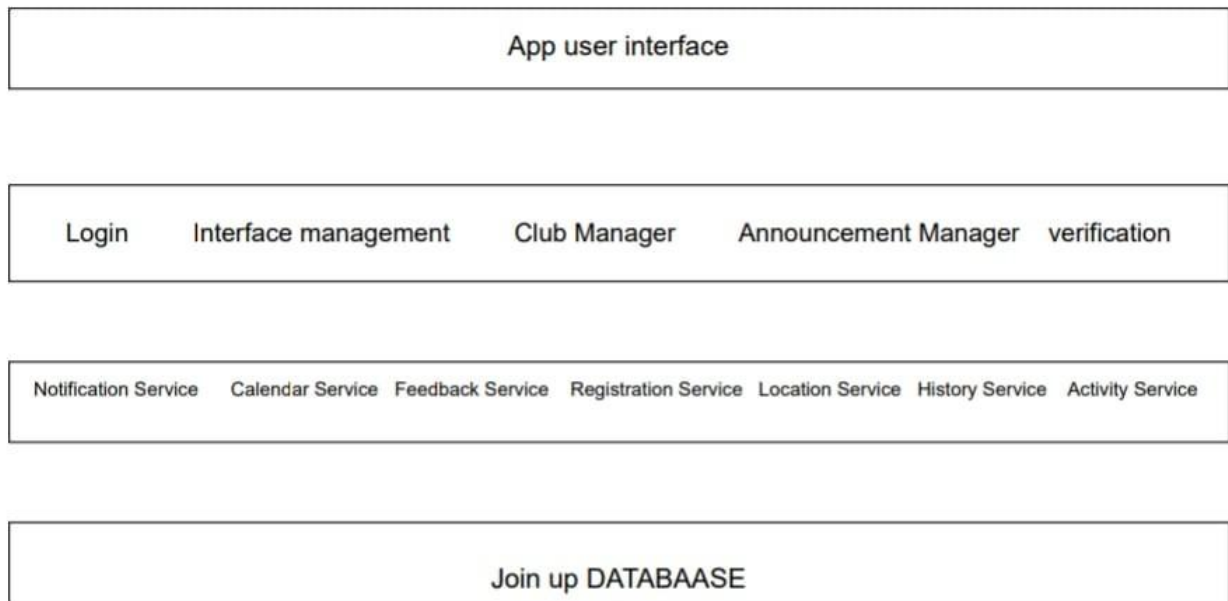
### Context Diagram:



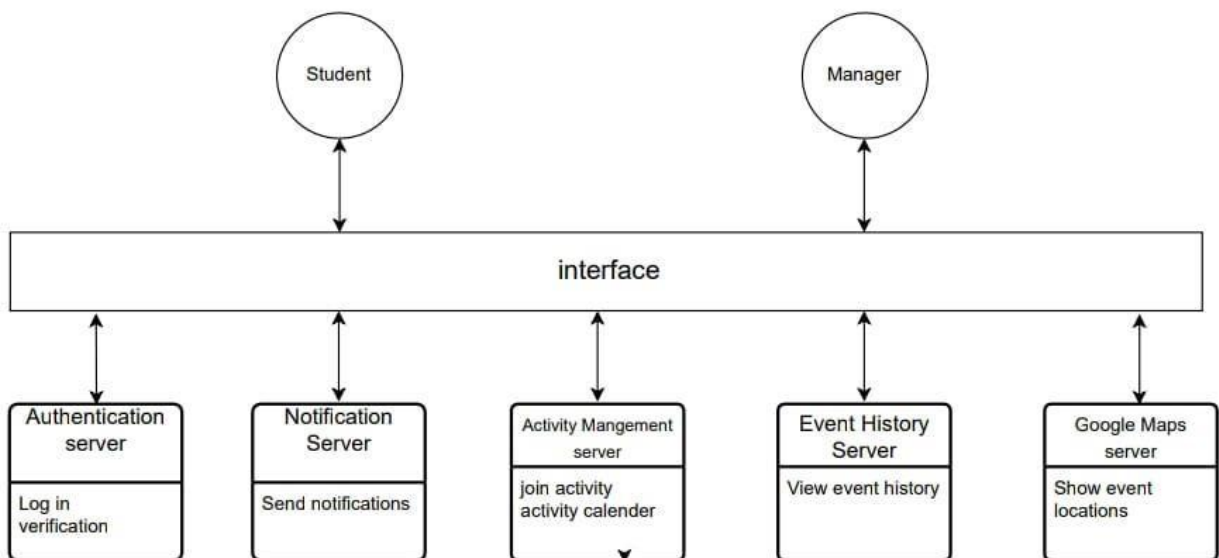
### User Case Diagram:



### Layered architecture:



### Client-server architecture:



## Use case scenario:

| 1. Login to Account |                                                                                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actors              | Student, Manager                                                                                                                                               |
| Description         | The user enters their email and password to log in. If the input is incorrect or does not match any account, they receive an error message and must try again. |
| Data                | Email and password                                                                                                                                             |
| Stimulus            | User submits login request.                                                                                                                                    |
| Response            | System validates the credentials, grants access if valid, or displays an error if invalid.                                                                     |
| Comments            | System should ensure proper formatting and encryption of credentials.                                                                                          |

| 2. Register for Activities |                                                                                                                                                                                                                           |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actors                     | Student                                                                                                                                                                                                                   |
| Description                | A student selects an available activity and submits a registration request. If the activity has open spots and meets the student's eligibility, they are successfully registered. Otherwise, they receive a notification. |
| Data                       | User ID, activity details                                                                                                                                                                                                 |
| Stimulus                   | User selects an activity and submits registration.                                                                                                                                                                        |
| Response                   | Enrolls the user in the activity and confirms the registration.                                                                                                                                                           |
| Comments                   | The system must verify the activity's availability before registration.                                                                                                                                                   |

| 3. View Personalized Calendar |                                                                                                                                                                          |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actors                        | Student                                                                                                                                                                  |
| Description                   | The student views a calendar displaying their registered activities with dates, times, and locations. Any updates to events are automatically reflected in the calendar. |
| Data                          | User ID, registered activity details                                                                                                                                     |
| Stimulus                      | The user requests to view their calendar.                                                                                                                                |
| Response                      | Retrieves and displays the student's registered activities in a calendar format.                                                                                         |
| Comments                      | The calendar should be user-friendly, support filtering options, and update dynamically to reflect any schedule changes.                                                 |

| 4. Receive Notifications |                                                                                                                                                                               |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actors</b>            | Student                                                                                                                                                                       |
| <b>Description</b>       | The student receives alerts about activity updates, schedule changes, or new events. Notifications are sent based on user preferences, either via email, app alerts, or both. |
| <b>Data</b>              | Notification preferences, event updates                                                                                                                                       |
| <b>Stimulus</b>          | Detects an update or event.                                                                                                                                                   |
| <b>Response</b>          | Notifications are sent via email, app alerts, or both based on user preferences.                                                                                              |
| <b>Comments</b>          | The system should ensure reliable delivery and allow notification customization.                                                                                              |

| 5. Manage Announcements |                                                                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actors</b>           | Manager                                                                                                                                            |
| <b>Description</b>      | A manager creates and schedules announcements for students. Announcements can include formatted text, attachments, and scheduled publishing times. |
| <b>Data</b>             | Announcement content, scheduling details                                                                                                           |
| <b>Stimulus</b>         | Manager submits an announcement creation request.                                                                                                  |
| <b>Response</b>         | Publishes the announcement and logs it.                                                                                                            |
| <b>Comments</b>         | Announcements should support formatting and scheduling features.                                                                                   |

| 6. Provide Feedback on Events |                                                                                                                                                               |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actors</b>                 | Student                                                                                                                                                       |
| <b>Description</b>            | A student submits feedback on events they attended, rating their experience and adding comments if needed. Only participants of an event can submit feedback. |
| <b>Data</b>                   | Feedback details (rating, comments)                                                                                                                           |
| <b>Stimulus</b>               | User submits feedback for an event.                                                                                                                           |
| <b>Response</b>               | Stores the feedback and confirms submission.                                                                                                                  |
| <b>Comments</b>               | Feedback should only be allowed for events the user attended.                                                                                                 |



| 7. View Event History |                                                                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actors</b>         | Student                                                                                                                                                       |
| <b>Description</b>    | The student reviews a list of past attended events, including participation details and submitted feedback. The history may include statistics and summaries. |
| <b>Data</b>           | User ID, event participation details                                                                                                                          |
| <b>Stimulus</b>       | User requests to view history.                                                                                                                                |
| <b>Response</b>       | Retrieves and displays event history with statistics.                                                                                                         |
| <b>Comments</b>       | Include visual representations like charts for better clarity.                                                                                                |

| 8. Display Event Locations with Google Maps |                                                                                                                    |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Actors</b>                               | Student, Google Maps                                                                                               |
| <b>Description</b>                          | The student selects an event to view its location on Google Maps, along with navigation options and venue details. |
| <b>Data</b>                                 | Event location details                                                                                             |
| <b>Stimulus</b>                             | User selects an event location.                                                                                    |
| <b>Response</b>                             | Displays the location and directions using Google Maps.                                                            |
| <b>Comments</b>                             | Integration with Google Maps must be seamless and accurate.                                                        |

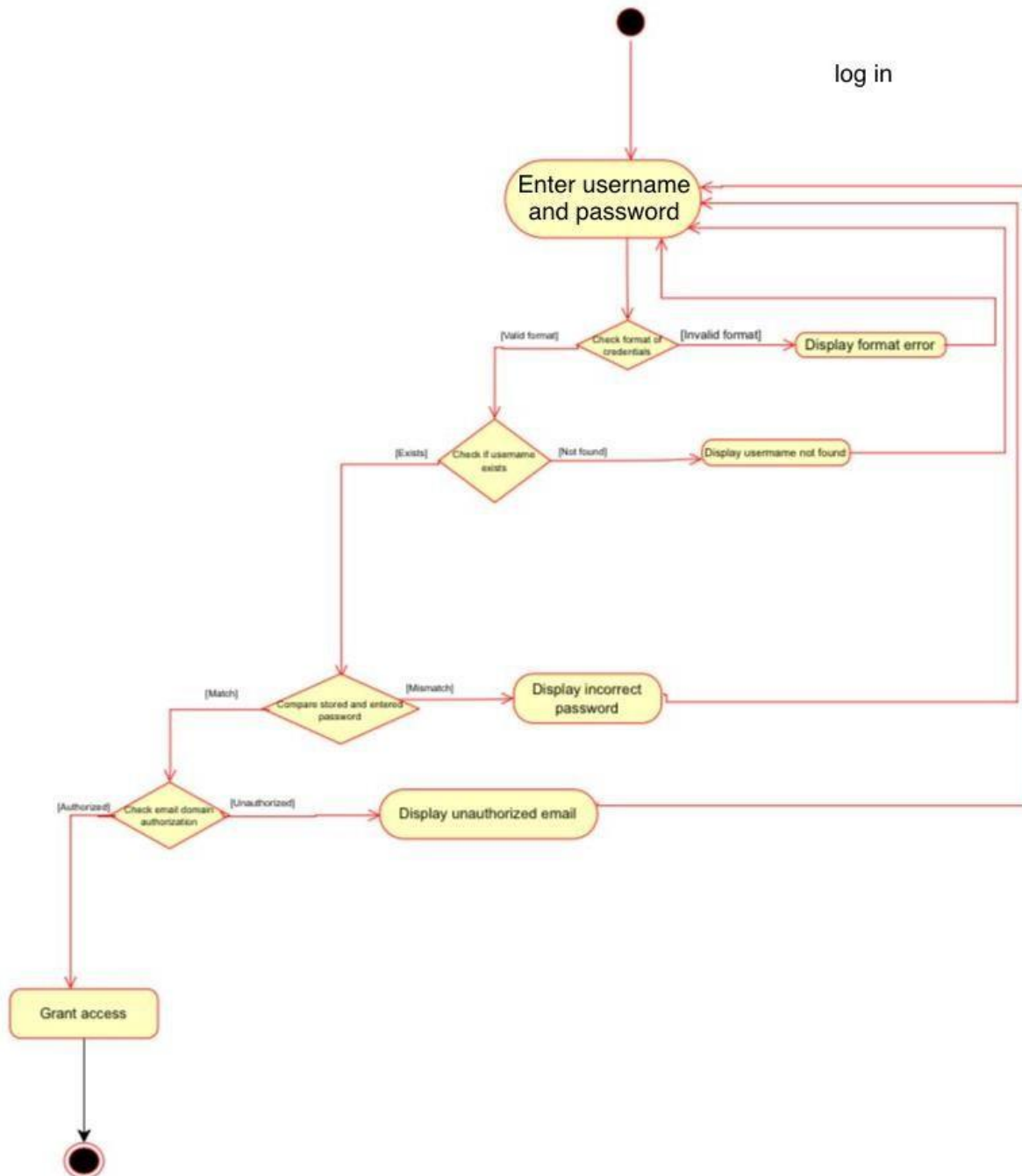
| 9. Cancel Activity Registration |                                                                                                                                              |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actors</b>                   | Student                                                                                                                                      |
| <b>Description</b>              | A student cancels their registration for an activity if permitted. The cancellation updates their calendar and the event's participant list. |
| <b>Data</b>                     | User ID, activity details                                                                                                                    |
| <b>Stimulus</b>                 | User submits a cancellation request.                                                                                                         |
| <b>Response</b>                 | Updates the database and confirms the cancellation.                                                                                          |
| <b>Comments</b>                 | Ensure that cancellations are reflected in the calendar and participant list.                                                                |

| 10. Register as Club Member |                                                                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actors</b>               | Student                                                                                                                                          |
| <b>Description</b>          | A student fills out a form with personal details to join a club. After submission, they receive confirmation and can update their profile later. |
| <b>Data</b>                 | Personal details (email, password, etc.)                                                                                                         |
| <b>Stimulus</b>             | User submits club membership registration.                                                                                                       |
| <b>Response</b>             | Validates, registers the user, and sends a confirmation email.                                                                                   |
| <b>Comments</b>             | Ensure secure data handling and allow profile updates after registration.                                                                        |

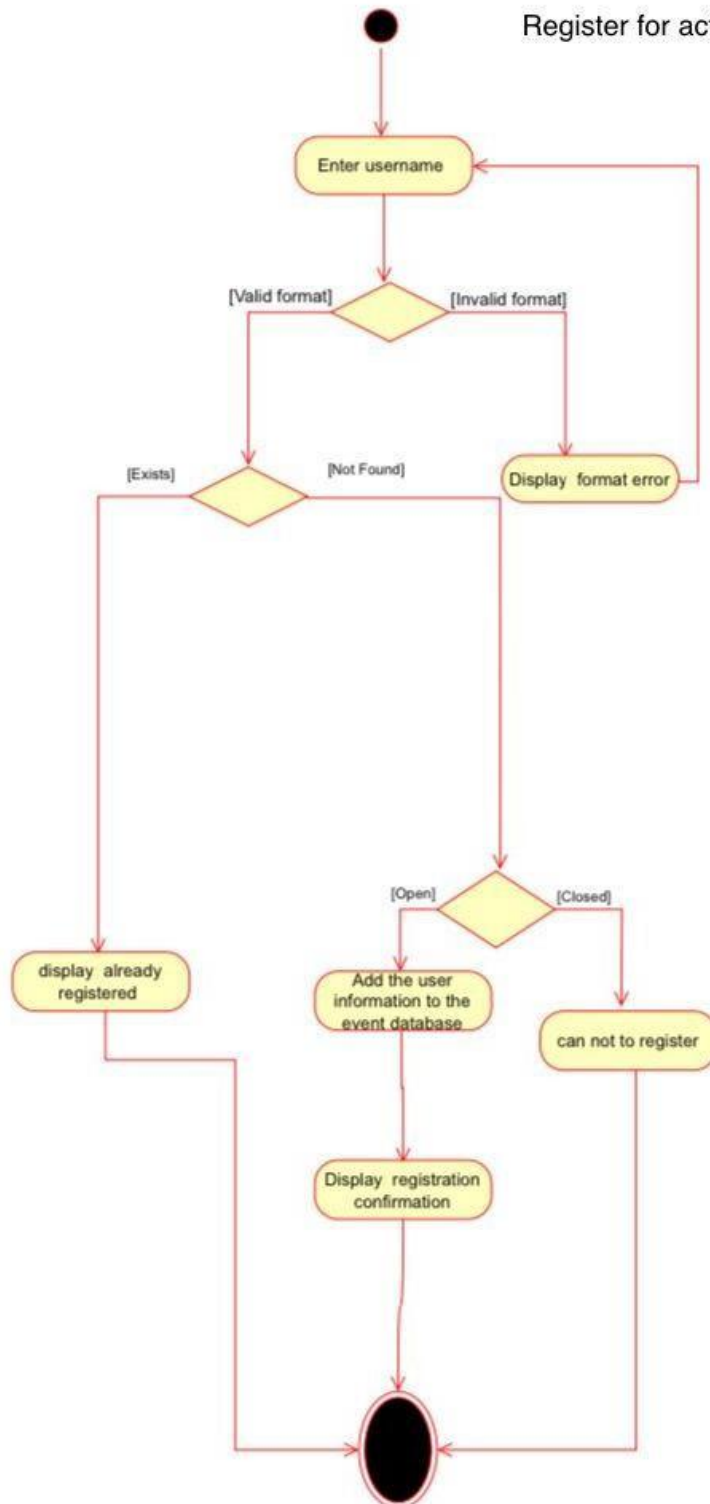
| 11. Email Verification After Logging In |                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actors</b>                           | Student, Manager                                                                                                                                                                                                                                                                                                                           |
| <b>Description</b>                      | After a user logs in, the system requires email verification for account security. It sends a verification code to the user's registered email, prompts the user to enter the code, and validates it. If the code is correct, the user gains full access. If incorrect, the system displays an error message and allows the user to retry. |
| <b>Data</b>                             | email, Verification Code, Error and Success Message                                                                                                                                                                                                                                                                                        |
| <b>Stimulus</b>                         | The user successfully logs into their account.                                                                                                                                                                                                                                                                                             |
| <b>Response</b>                         | The system sends a verification code to the user's email, prompts them to enter it, validates the code, grants access if correct, or displays an error and allows retry if incorrect.                                                                                                                                                      |
| <b>Comments</b>                         | The system should allow the user to request a new verification code if they do not receive the original email or if the code expires.                                                                                                                                                                                                      |

## Chapter 6 (Software Modeling):

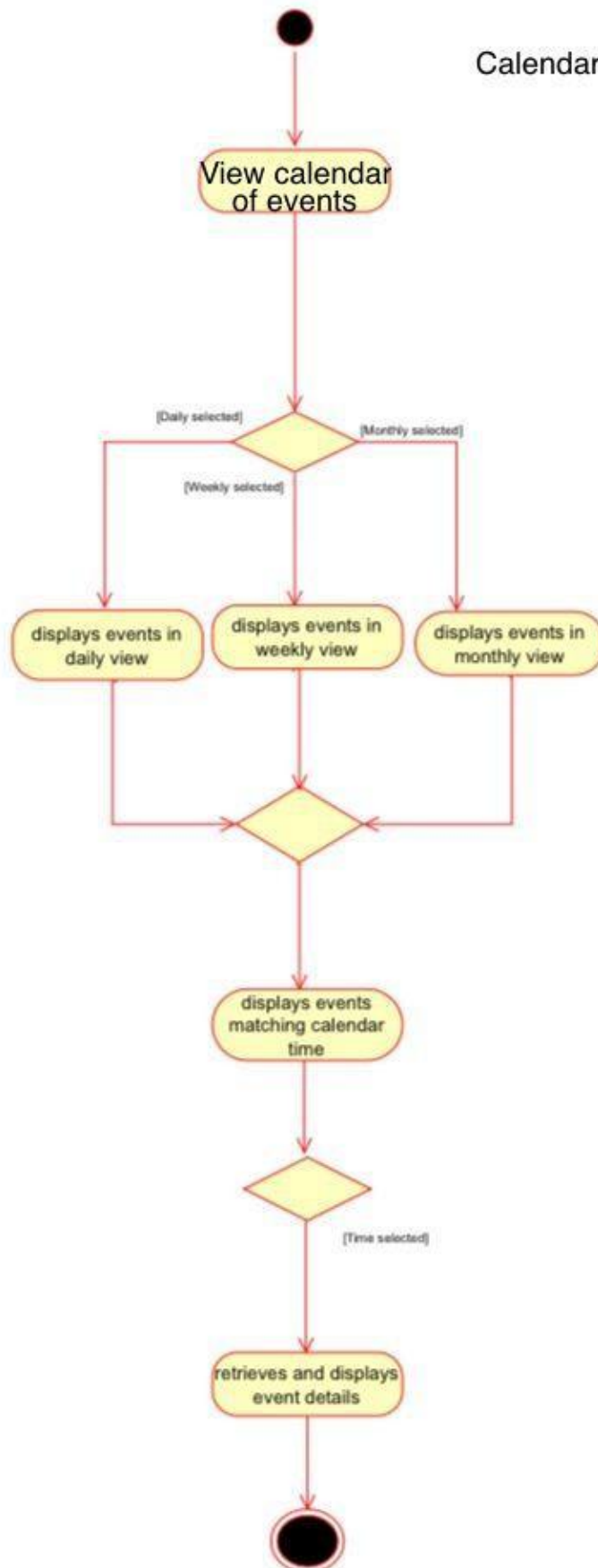
Active diagrams:



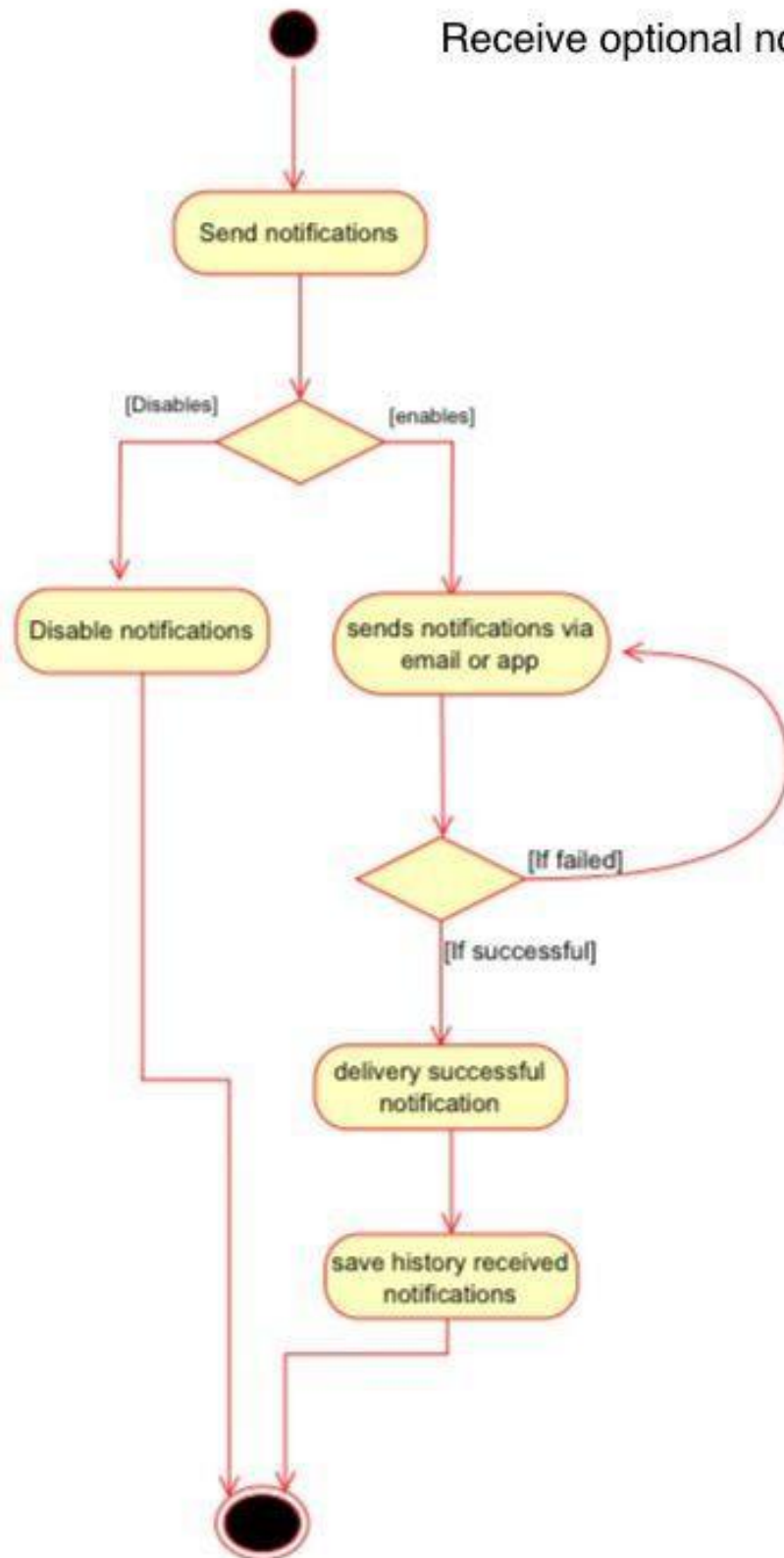
## Register for activities



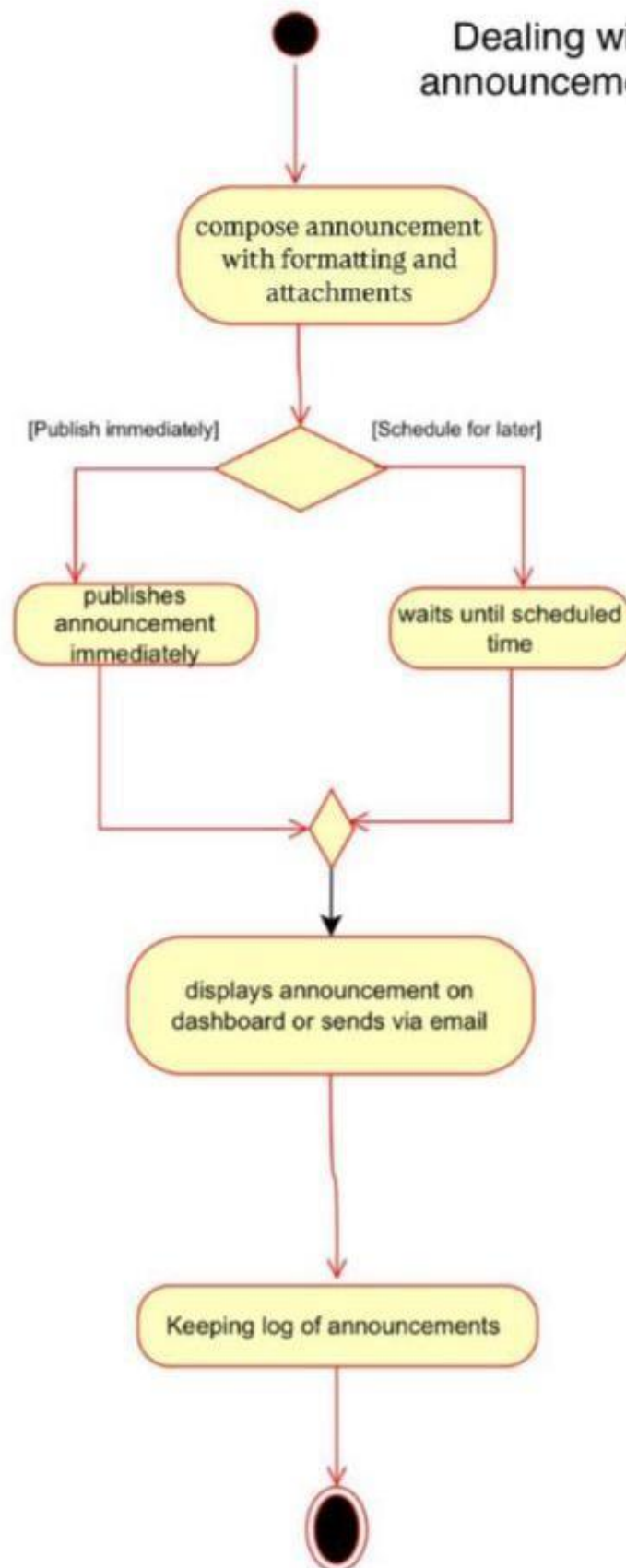
## Calendar of recorded activities



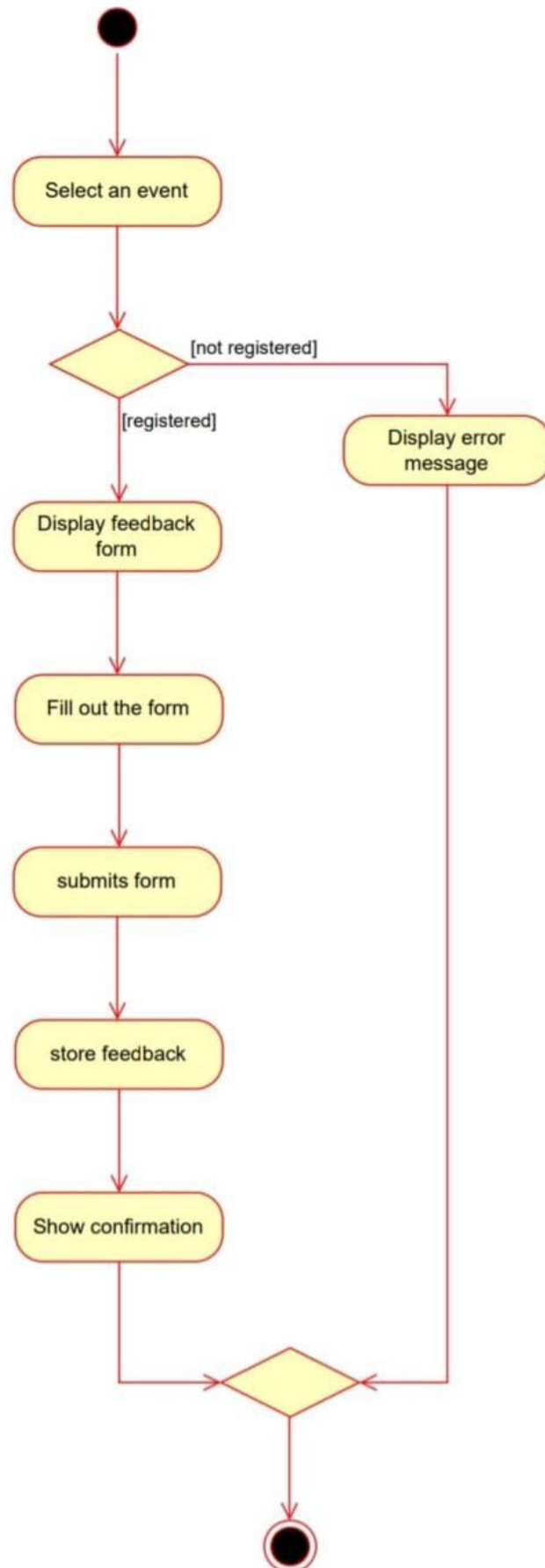
## Receive optional notifications



## Dealing with announcements

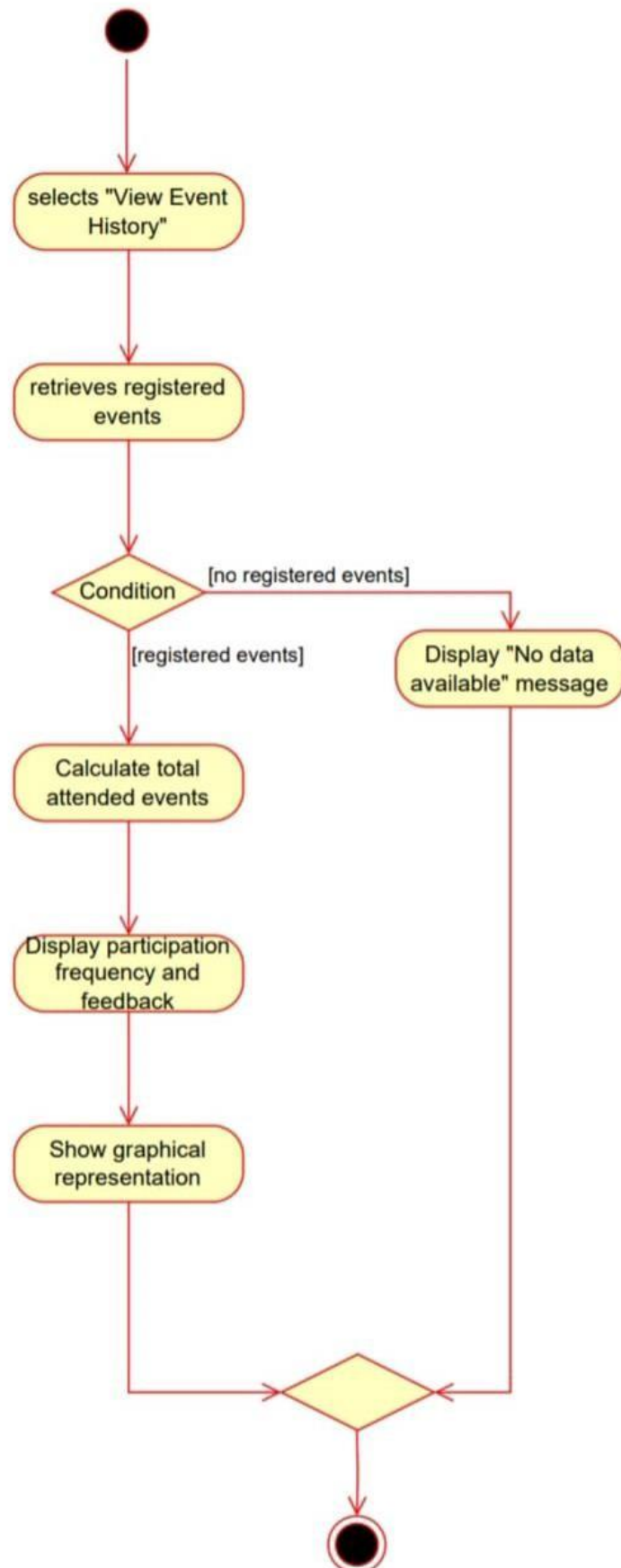


# User Feedback on Events

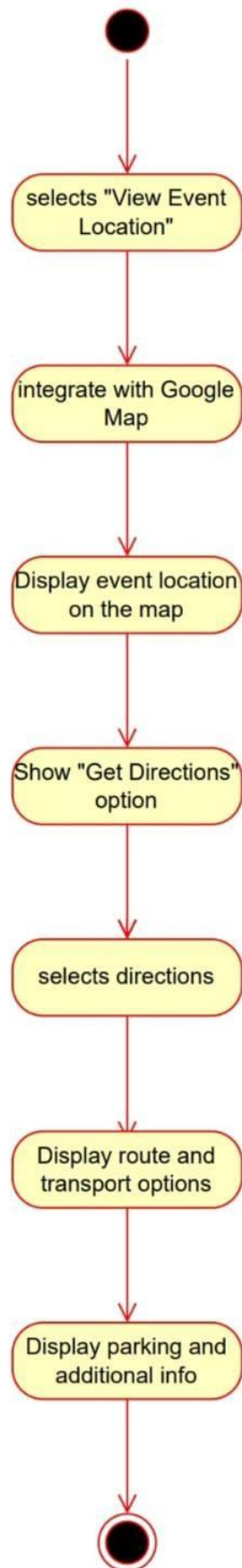




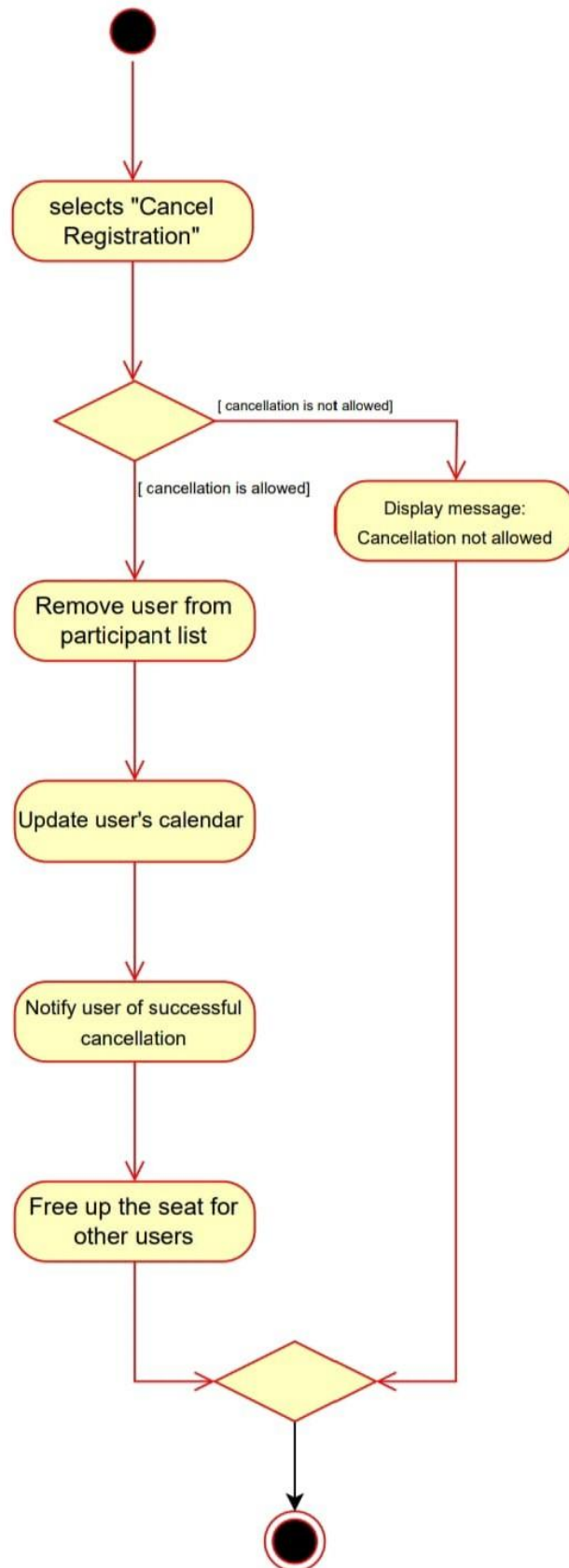
## Event History



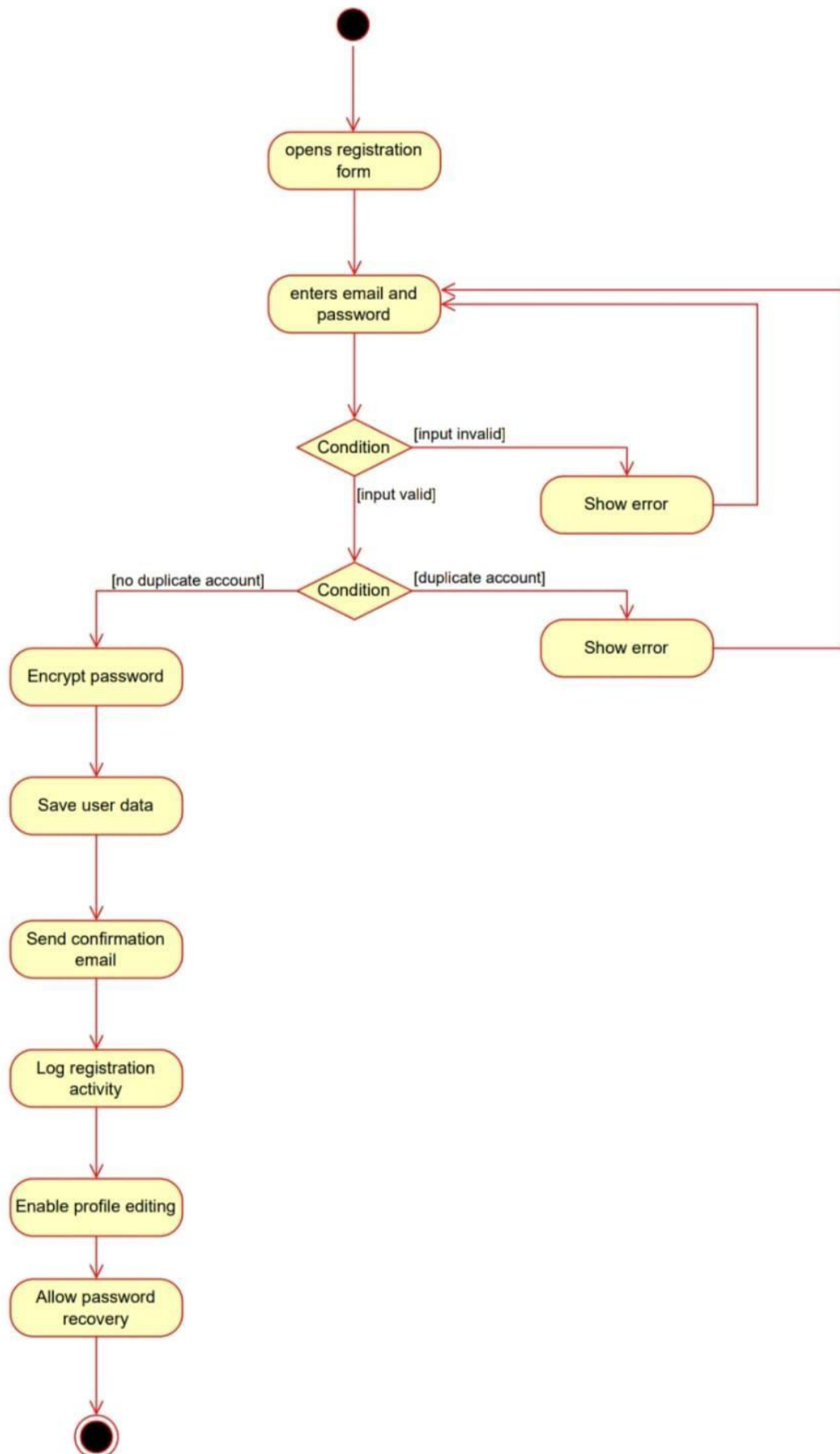
event locations



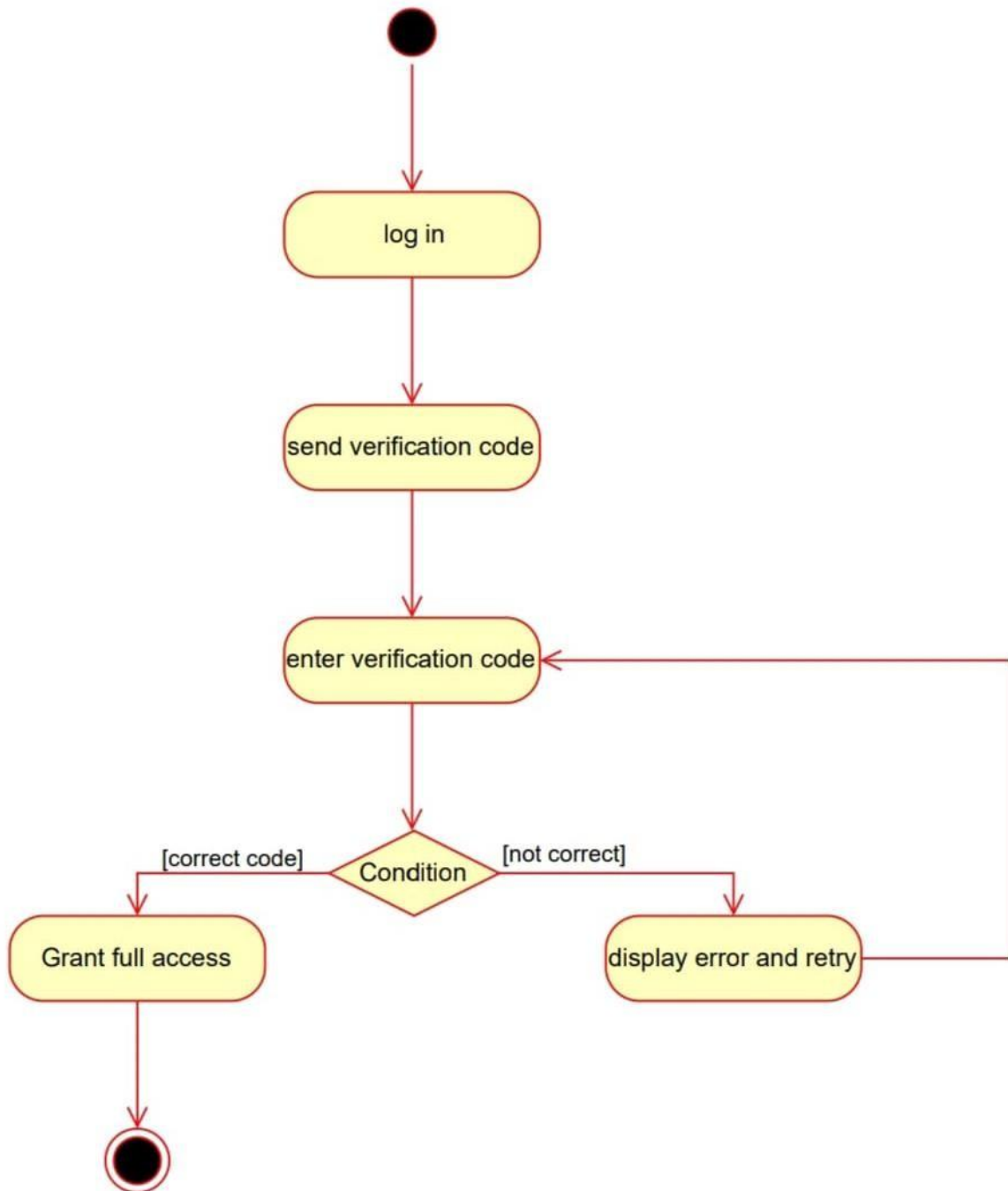
cancel activity registrations



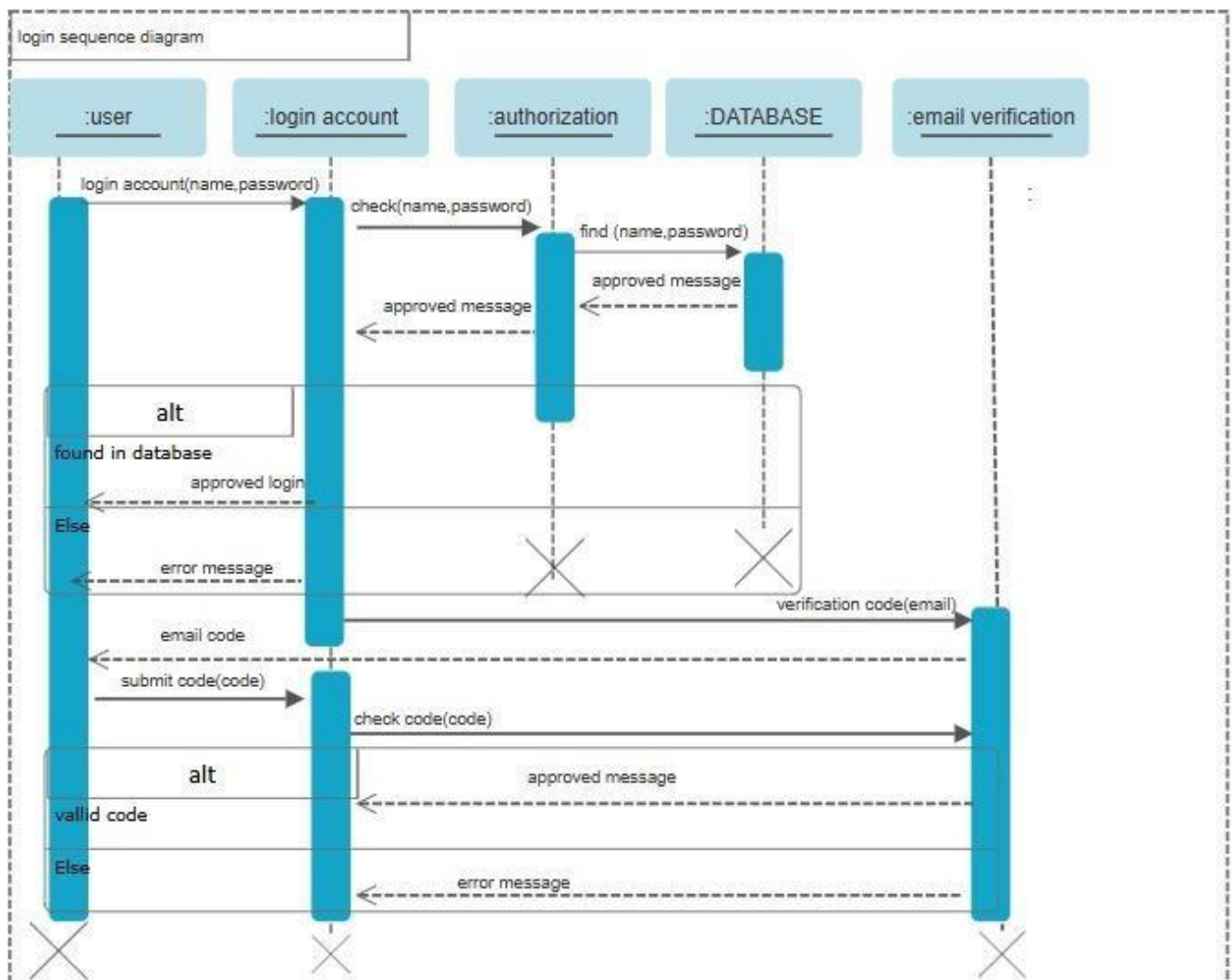
register as a club member

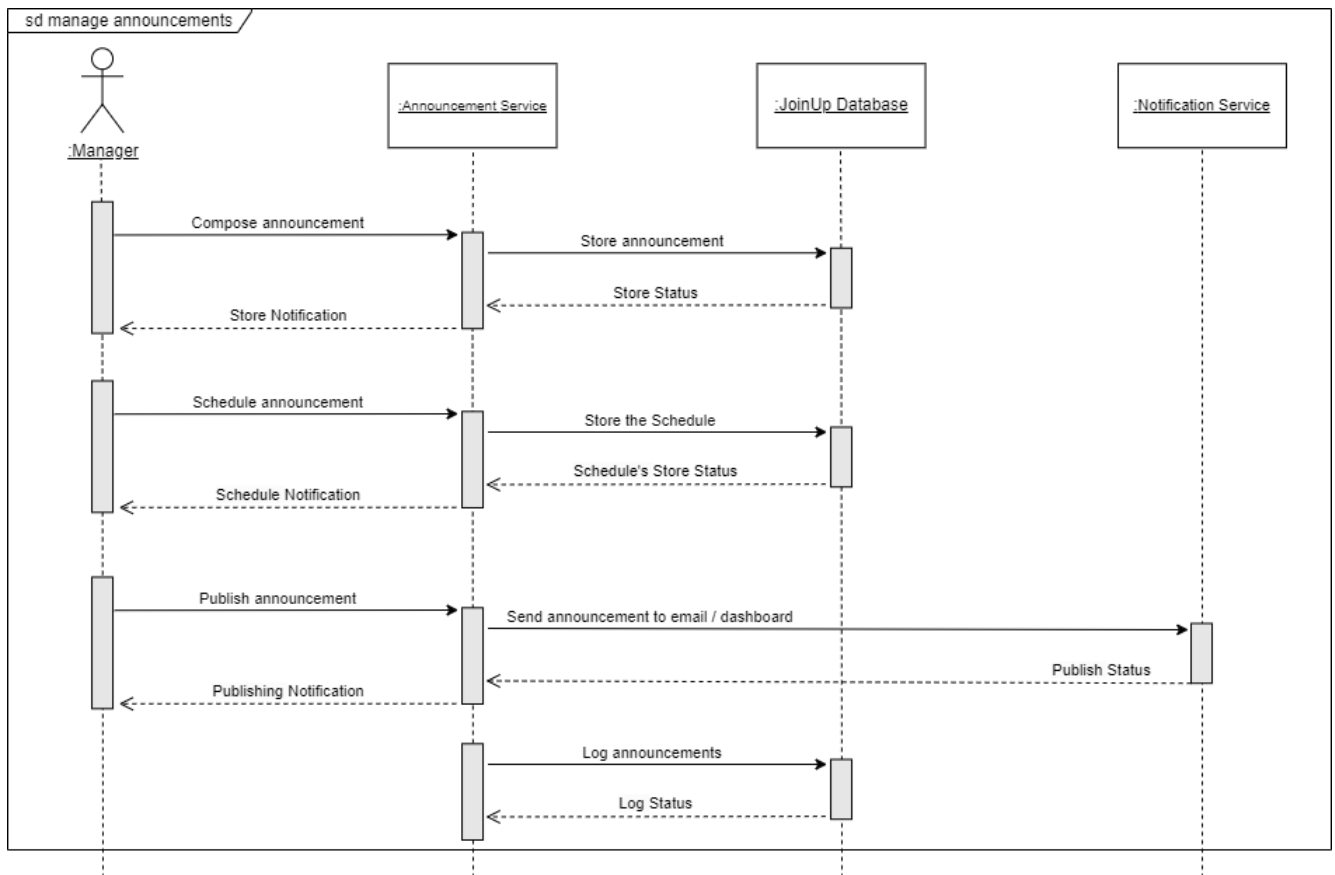


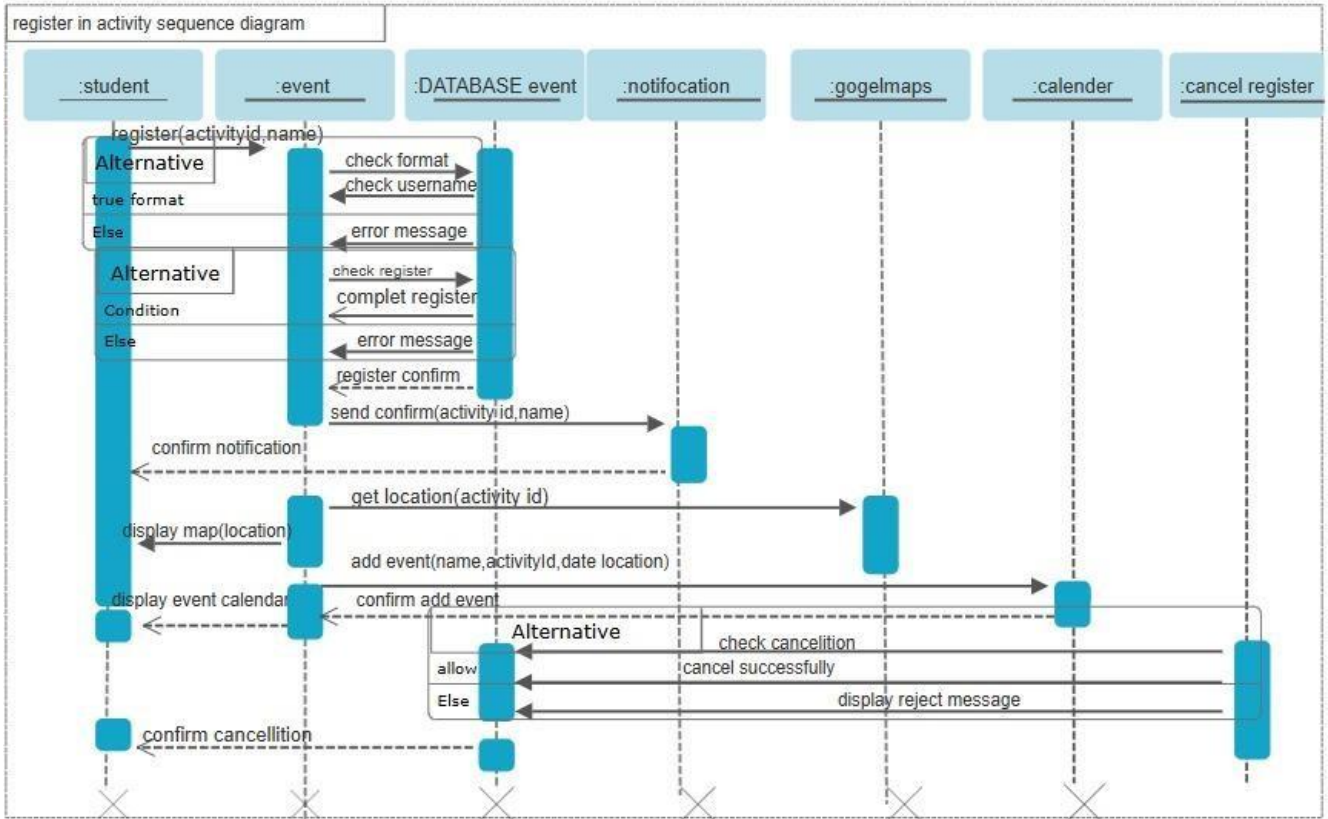
verification after logging in



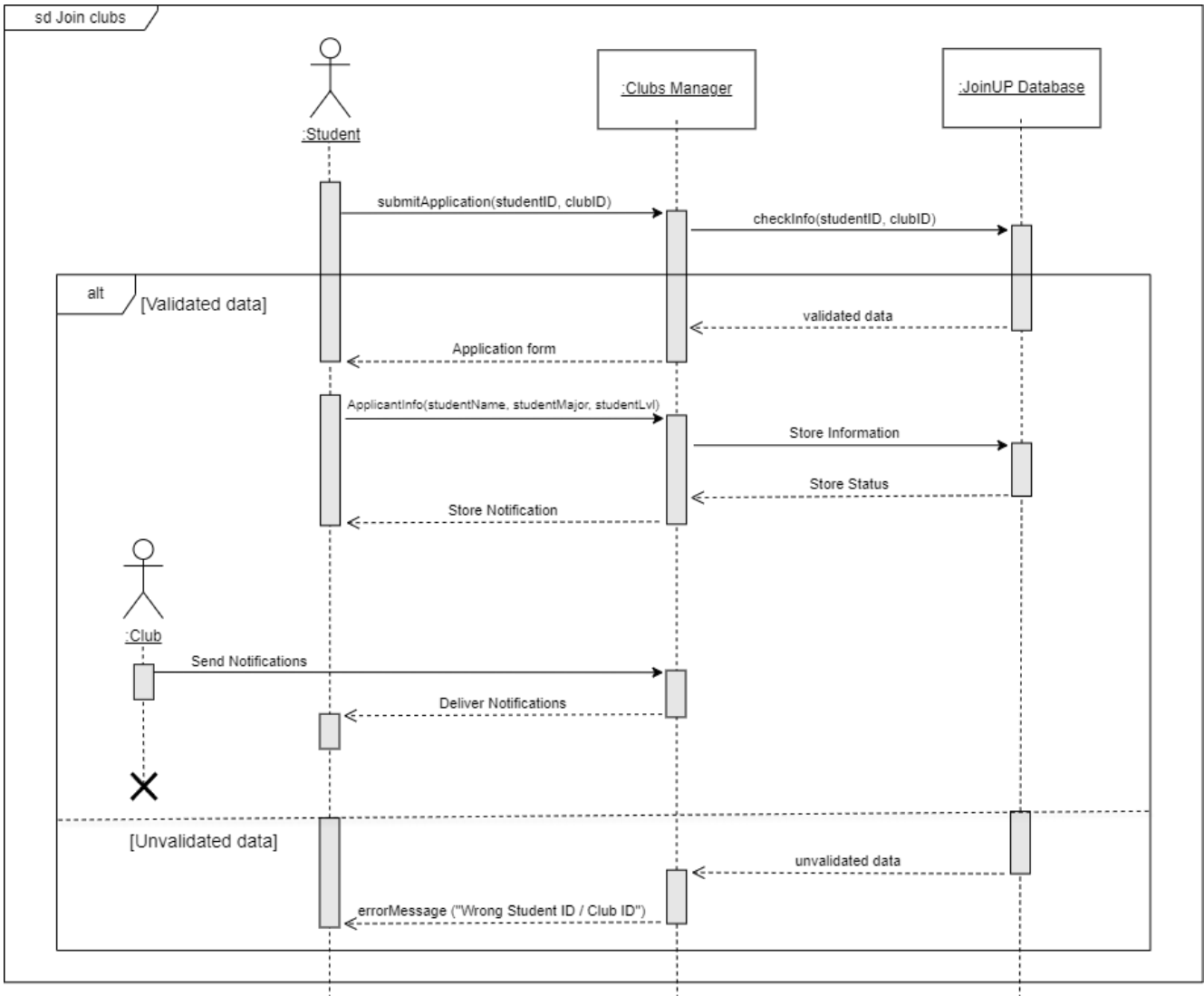
## Sequence diagrams:



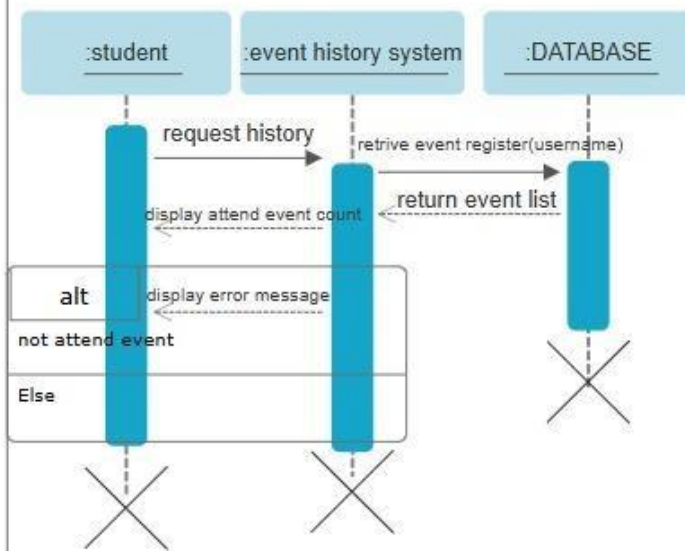


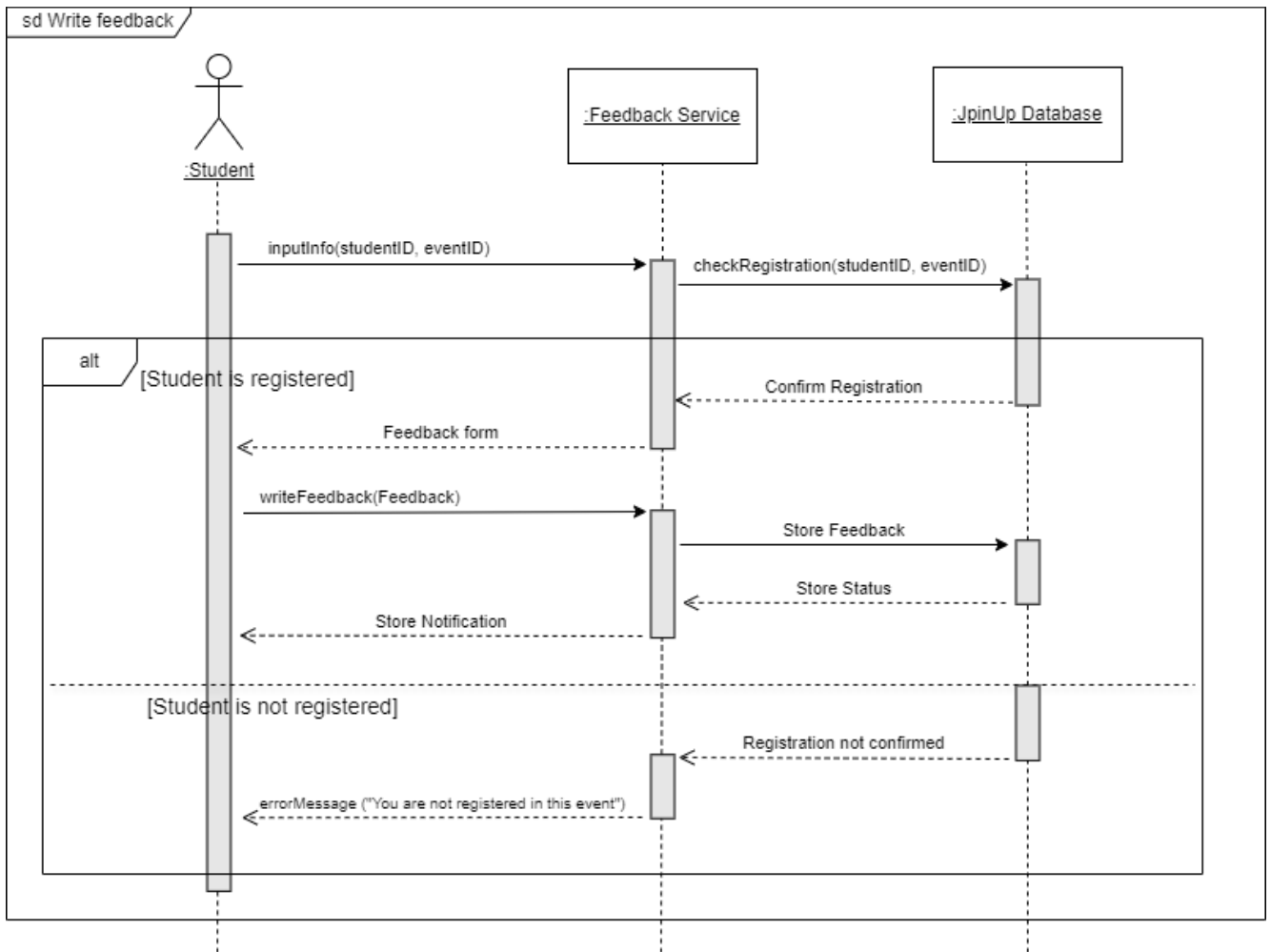




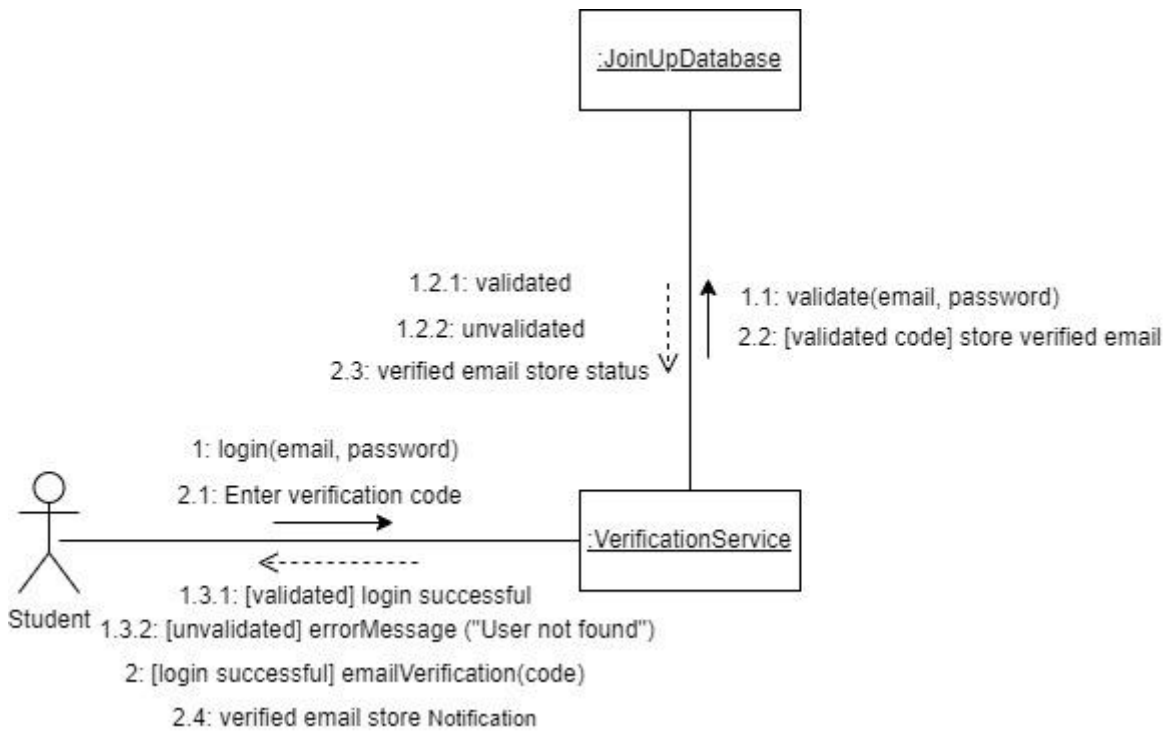


event history sequence diagram

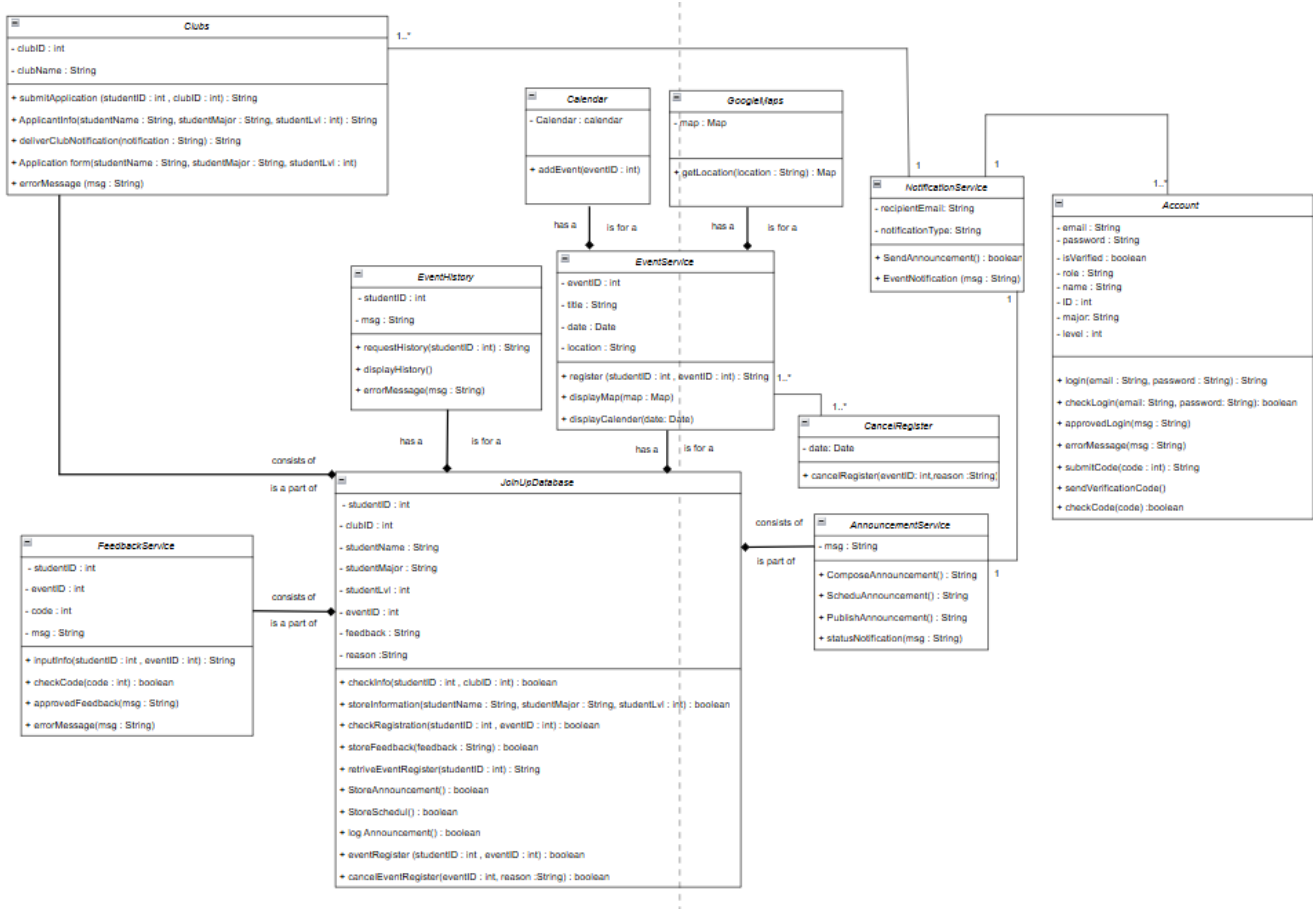




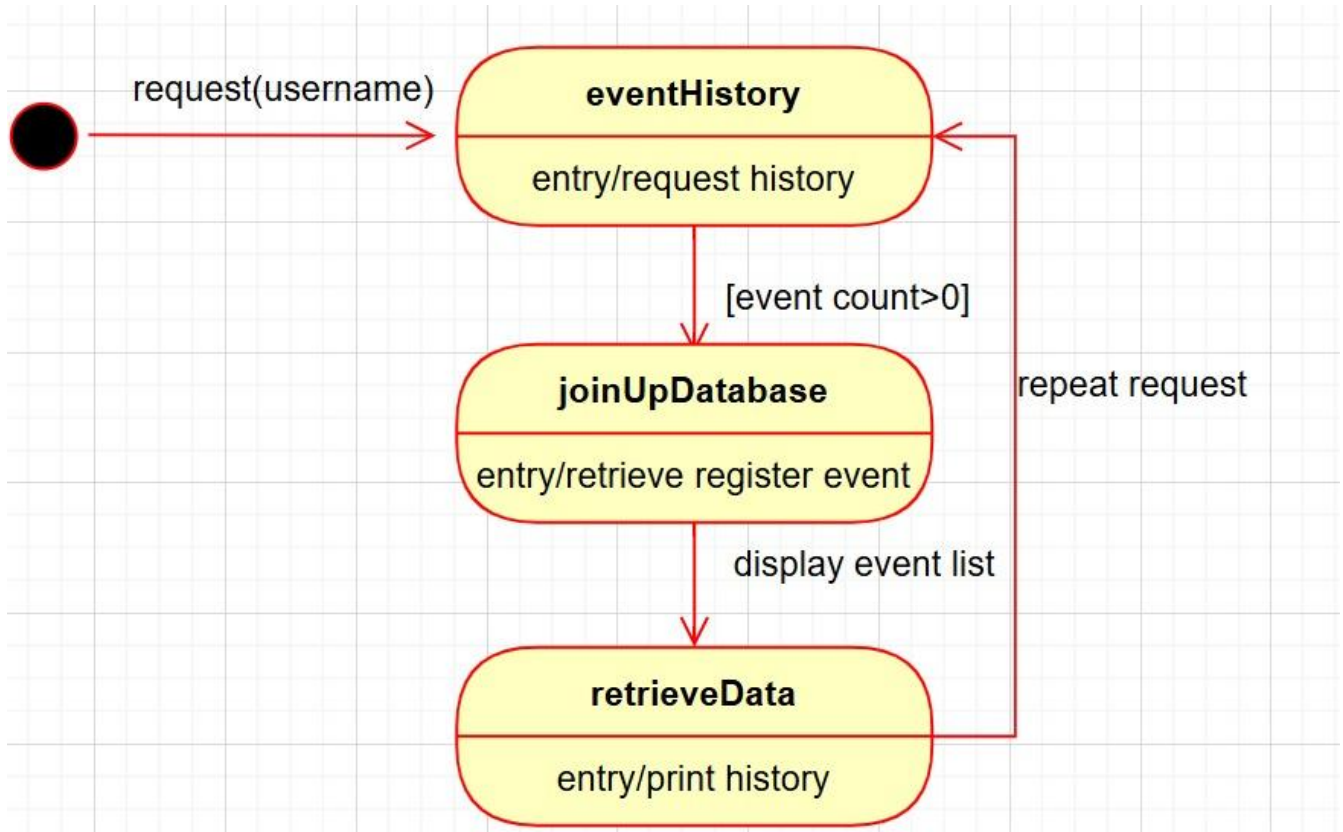
Communication Diagram:



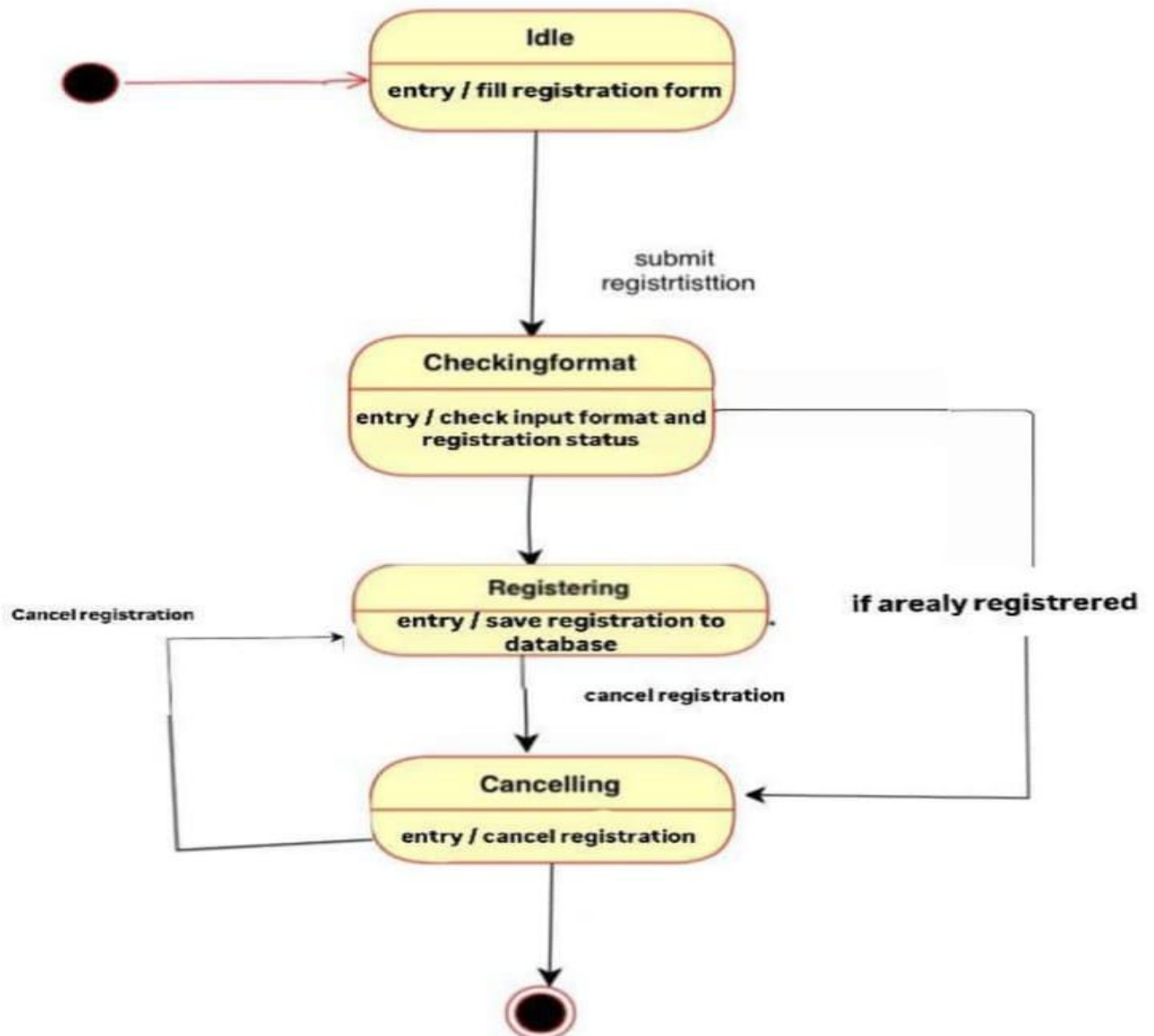
Class Diagram:



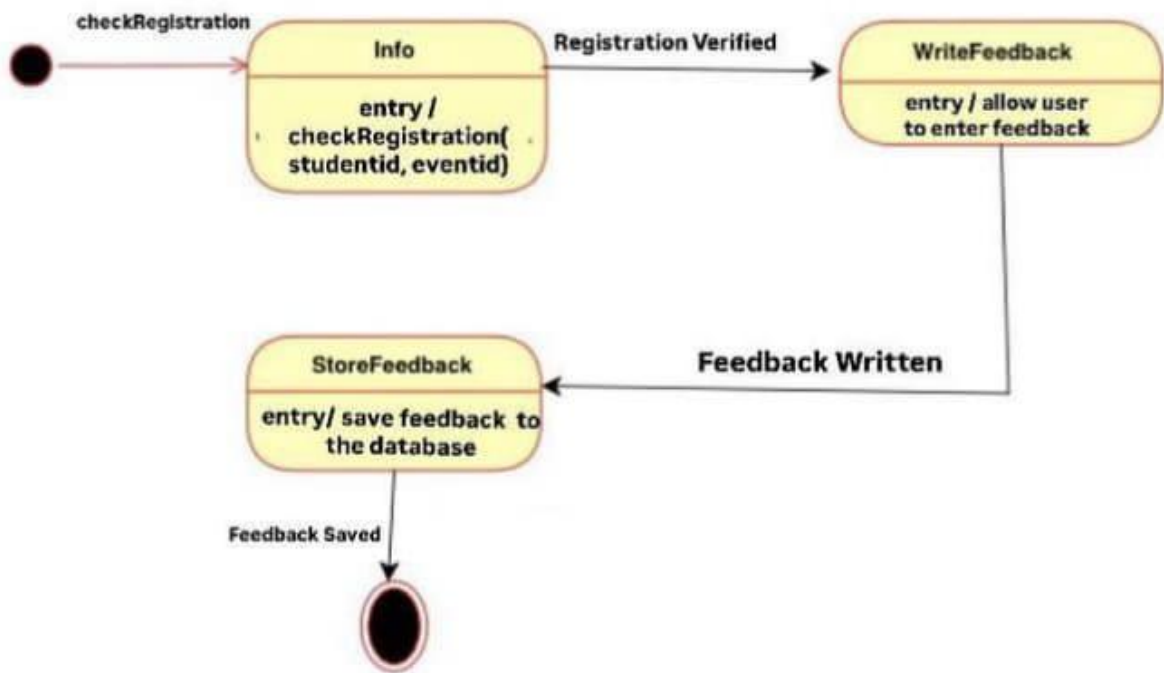
**Stat Diagram:**



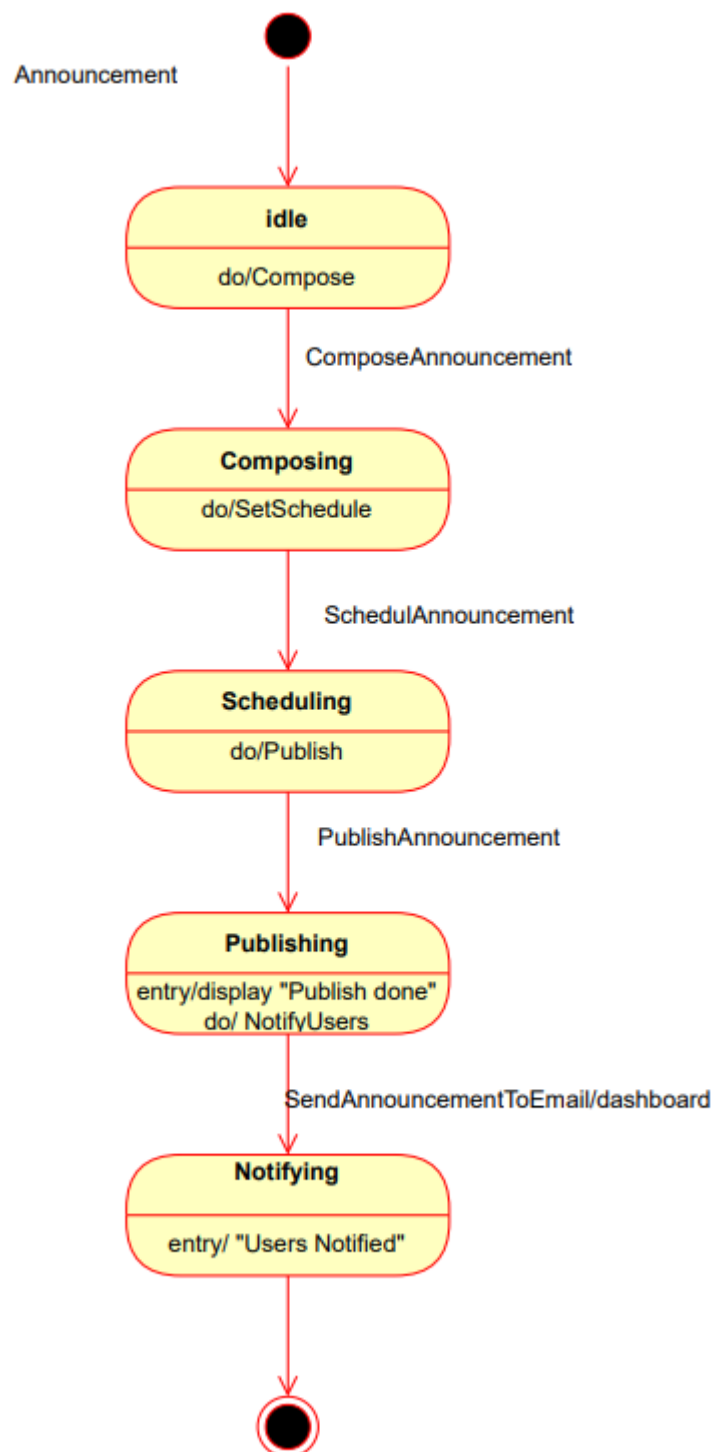
## Student registration



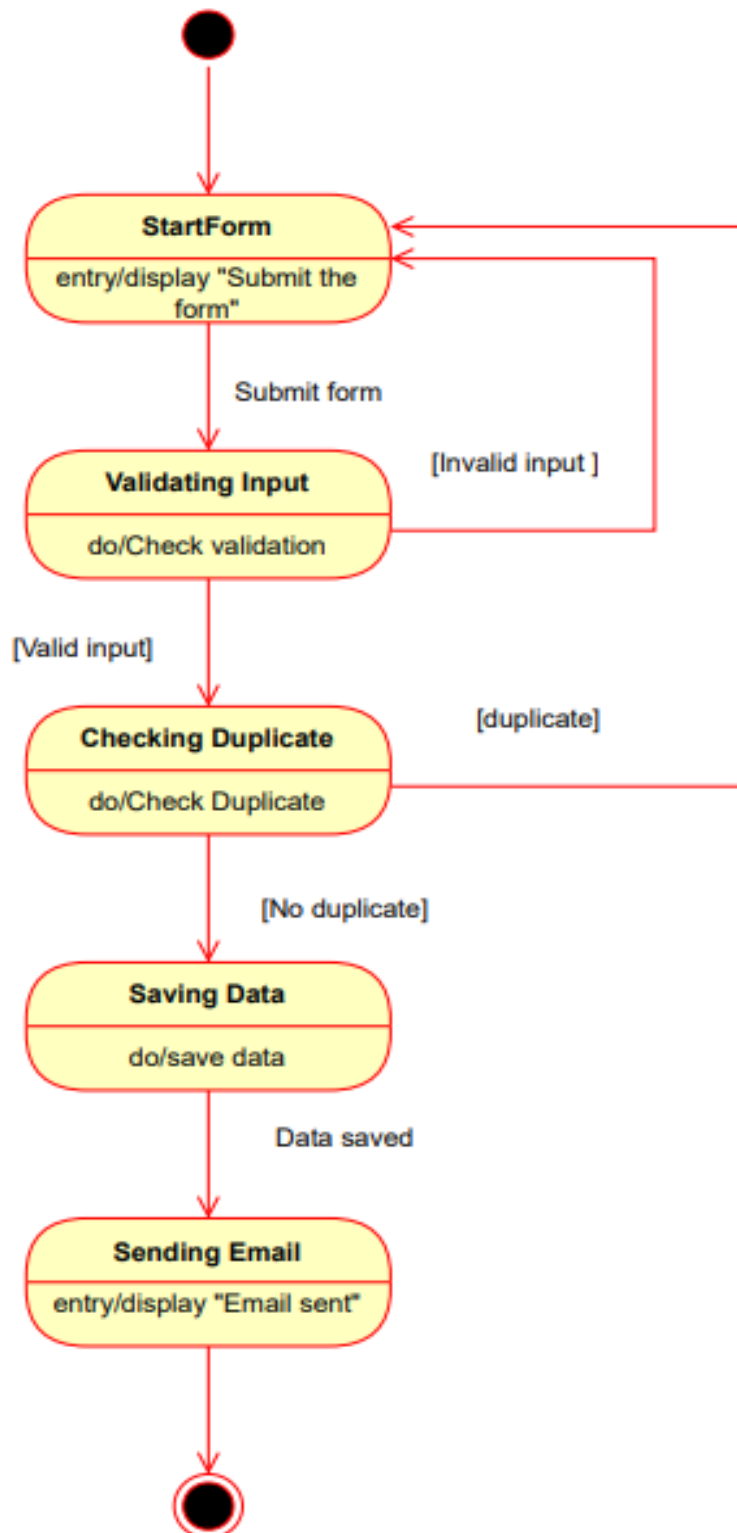
## Write feedback



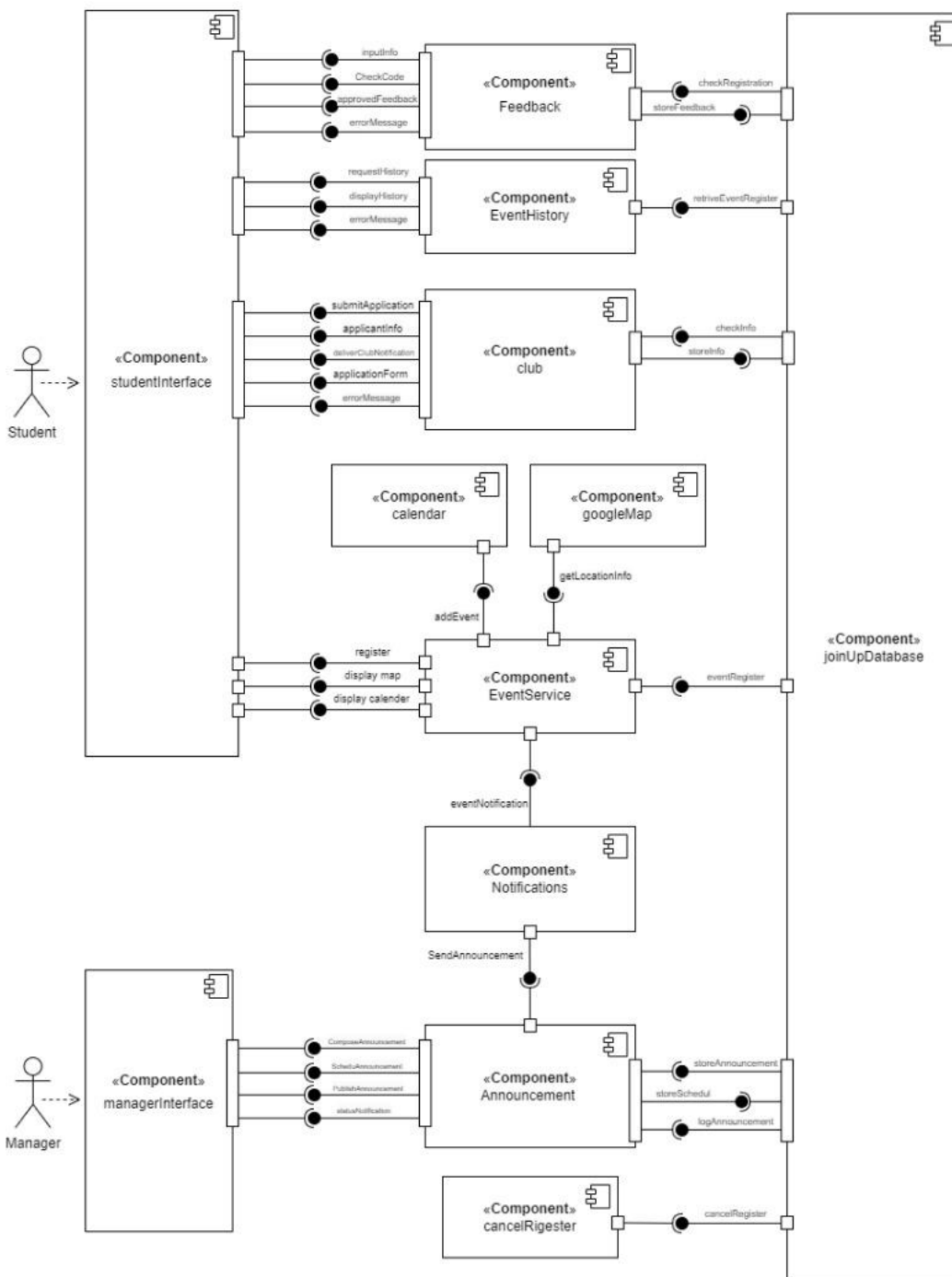
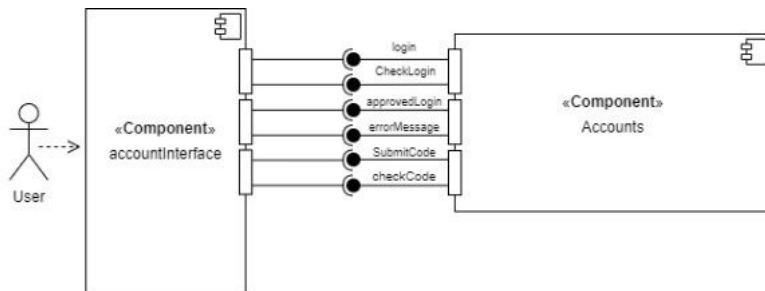




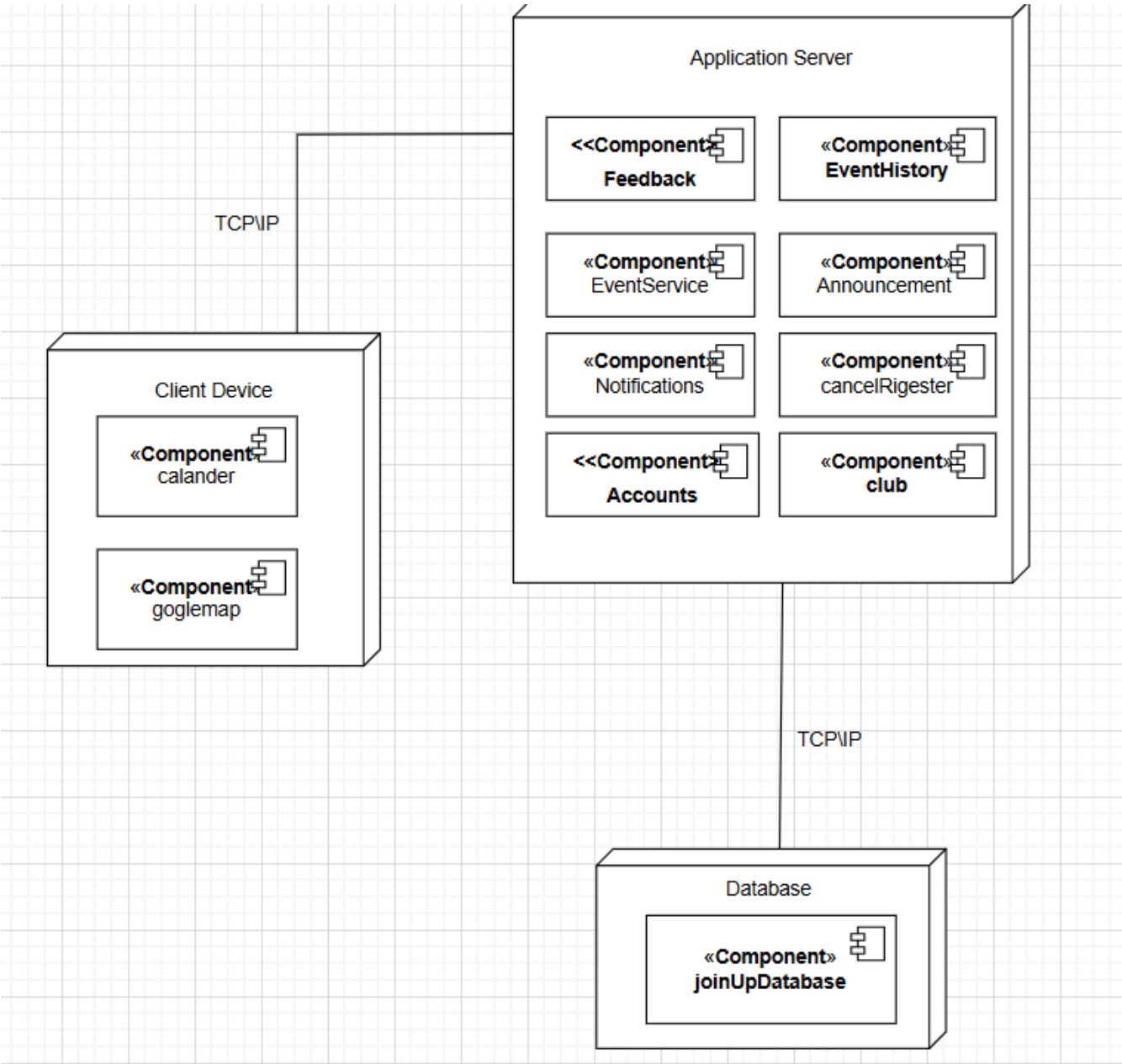
join a club



**Component diagrams:**



Deployment diagrams:



Meeting Minutes:

| Members who attended the meeting                                 | Date and Time                          |
|------------------------------------------------------------------|----------------------------------------|
| Renad Al-Harbi<br>WED BADAHDAH<br>TAHANI ALBARQI<br>SAWSAN OMAR  | 8\3\2025 2:00 – 3:00 PM                |
| Renad Al-Harbi<br>WED BADAHDAH TAHANI<br>ALBARQI<br>SAWSAN OMAR  | 14\3\2025<br>15\3\2025<br>16\3\2025    |
| Renad Al-Harbi<br>Wed Badahdah<br>Tahani Albariqi<br>Sawsan Omar | May 1–4, 2025   2:00 PM – 11:00 PM     |
| Renad Al-Harbi<br>Wed Badahdah<br>Tahani Albariqi<br>Sawsan Omar | April 11–13, 2025   2:00 PM – 11:00 PM |

**Tasks:**

- REVIEW CHAPTER 4 – RENAD ALHARBI
- REVIEW CHAPTER 1.2 – TAHANI ALBARQI
- REVIEW CHAPTER 3 – WED BADAHDAH
- REVIEW CHAPTER 5 – SAWSAN OMAR
  
- **Sequence diagrams** – RENAD ALHARBI
- **Active diagrams**– TAHANI ALBARQI
- **Sequence diagrams+ Communication Diagram** – WED BADAHDAH
- **Active diagrams**– SAWSAN OMAR
  
- **Class Diagram**- Tahani ALBARIQI , WED BADAHDAH , SAWSAN OMAR ,RENAD ALHARBI
- **State Diagrams**- Tahani ALBARIQI, SAWSAN OMAR, RENAD ALHARBI
- **Component Diagrams** - Tahani ALBARIQI , WED BADAHDAH , SAWSAN OMAR ,RENAD ALHARBI
- **Deployment Diagrams** - SAWSAN OMAR , Tahani ALBARIQI